Error-Corrected Heterogeneous Treatment Effects: When LLM Annotation Debiasing Hurts

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Abstract

We derive an exact bias identity for heterogeneous treatment effect (HTE) analysis with LLM-annotated effect modifiers: correcting annotation error with a pooled confusion matrix—standard practice—introduces subgroup-level bias proportional to $\delta_s \cdot \Delta\tau$, the product of each subgroup's misclassification deviation ($\delta_s = \pi_s - \bar{\pi}$) and CATE heterogeneity, verified at $R^2 \geq 0.97$ on both synthetic grids and 11 real LLM confusion matrices. The identity implies that global correction worsens bias in 50% of K=2 configurations across our synthetic parameter grid (260/520 $\Delta\pi \times \Delta\tau$ cells), with amplification up to 2.86× (difference-in-means) and 5.9× (K=4, CausalForest). Independently, the identity yields a computable diagnostic (AUC = 0.86 under symmetric errors) that flags harmful correction using only per-subgroup misclassification rates—actionable without estimating treatment effects. The identity prescribes per-subgroup correction; we provide a hierarchical Bayesian implementation (EC-HTE) that reduces |bias| from 0.059 to 0.010 under extreme misclassification heterogeneity (K=2) with nominal coverage (≥0.91), but whose advantage is conditional: when misclassification heterogeneity is moderate and subgroups are unbalanced (e.g., TweetEval), EC-HTE yields comparable bias but higher aggregate RMSE than global correction (0.225 vs. 0.124), confirming that the method requires sufficient per-subgroup error-rate heterogeneity. We validate across 520 synthetic configurations (100 Monte Carlo runs each), a semi-synthetic TweetEval experiment (N=5,000), and eight LLMs (open-source 7B–9B and proprietary) across two NLP datasets, confirming both $\Delta\pi$ heterogeneity and the bias identity ($R^2 = 0.975$) on real annotations; EC-HTE is robust to moderate outcome-dependent misclassification ($\gamma \leq 0.2$, coverage ≥93.5%).

1 Introduction

Large language models (LLMs) now routinely replace human annotators in computational social science (CSS), producing labels for sentiment, hate speech, stance, and political discourse at costs orders of magnitude below manual coding (Gilardi et al., 2023; Törnberg, 2025; Ziems et al., 2024). Yet these annotations are not uniformly accurate. Ashwin et al. (2025) document that LLM misclassification rates vary by up to 5× across demographic subgroups defined by refugee status, gender, and education level. In a pilot study, GPT-4.1 achieves 85.8% binary accuracy on TweetEval sentiment, but error rates range from 10.2% to 20.0% across text-feature subgroups ($\Delta\pi = 0.098$). When such annotations serve as effect modifiers in heterogeneous treatment effect (HTE) analysis, subgroup-specific errors propagate directly into conditional average treatment effect (CATE) estimates. The resulting bias is not random noise that averages out; it is structured by the same demographic axes that define the research question. Standard correction procedures[1], which estimate a single confusion matrix and apply its inverse uniformly, cannot account for this heterogeneity.

We formalize this problem and derive an exact bias identity for the correction-induced error. When a pooled confusion matrix is applied to correct subgroup CATEs, it misattributes error across subgroups with an exact product form: the bias in subgroup s equals $\delta_s \cdot \Delta\tau/(1 - 2\bar{\pi})$, where δ_s is the deviation from the pooled error rate and $\Delta\tau$ is the CATE heterogeneity (Theorem 1). This is not a bound but an exact population-level identity, verified empirically with $R^2 = 0.994$ (K=2) and $R^2 = 0.974$ (K=4) on synthetic grids, and $R^2 = 0.975$ across 11 real LLM confusion matrices spanning open-source and proprietary model

[1]Throughout, "debiasing" refers to correction of annotation measurement-error bias, not algorithmic fairness debiasing.

families. A direct consequence is that global correction can be worse than no correction at all: global correction worsens CATE bias in 50% of $K{=}2$ configurations (260/520) across our parameter grid—concentrated in regimes where $\Delta\pi > 0.10$ and per-subgroup expert samples exceed 25—with amplification reaching 2.86× ($K{=}2$) and 5.9× ($K{=}4$ with CausalForest). Error-Corrected Heterogeneous Treatment Effects (EC-HTE) addresses this through hierarchical Bayesian subgroup-stratified confusion matrix estimation, reducing |bias| from 0.059 to 0.010 under extreme heterogeneity ($K{=}2$). The bias decomposition identity also yields a $\Delta\tau$-free diagnostic (Proposition 3): a condition on per-subgroup misclassification rates alone predicts when global correction will increase bias (AUC = 0.86)—independently useful even without adopting EC-HTE, since practitioners can flag harmful correction from a pilot sample before estimating any treatment effects. EC-HTE's advantage is conditional on sufficient error heterogeneity and expert budget; when $\Delta\pi$ is small, the variance cost of stratification outweighs its bias benefit—the expected bias-variance tradeoff of any subgroup-stratified correction (Figure 1).

Subgroup CATEs inform resource allocation: whether a content moderation policy reduces toxicity more in long-form versus short-form posts, or whether a public health campaign reaches different demographic groups equally. Biased CATE estimates can reverse the sign of apparent treatment heterogeneity, leading practitioners to allocate resources away from communities that benefit most. The failure mode is subtle: a researcher who corrects for annotation noise using a pooled confusion matrix may introduce more bias than doing nothing, precisely in the low-error subgroups where original labels were most trustworthy.

Existing corrections—PPI (Angelopoulos et al., 2023), DSL (Egami et al., 2024), PPCI (Sui and Candès, 2026), arm-specific PPI (Mozer, 2026), ForestIV (Bu et al., 2023)—address population-level or outcome-side measurement error but assume homogeneous prediction error across subgroups; none corrects bias when the mismeasured variable is the effect modifier indexing the CATE function (Section 2).

Our contributions are:

- A bias decomposition identity (Theorem 1) proving that global correction introduces subgroup bias proportional to $\delta_s \cdot \Delta\tau$ (per-subgroup misclassification deviation × CATE heterogeneity), with a $\Delta\tau$-free diagnostic (Proposition 3; AUC = 0.86) that flags when correction is harmful using only per-subgroup misclassification rates—independently actionable without EC-HTE (Sections 3, 5.4).

- EC-HTE, the natural correction prescribed by the bias structure: per-subgroup confusion matrix estimation via hierarchical Bayes, compatible with a range of CATE estimators and composable with PPCI for dual covariate/outcome error (|bias|: 0.179 → 0.007; Sections 4–5.4).

- Applicability conditions and crossover analysis ($n_{\text{expert}}{=}250$ for $K{=}2$, $n_{\text{expert}}{=}1{,}000$ for $K{=}4$) establishing when EC-HTE improves over global correction and when it does not (Appendix I).

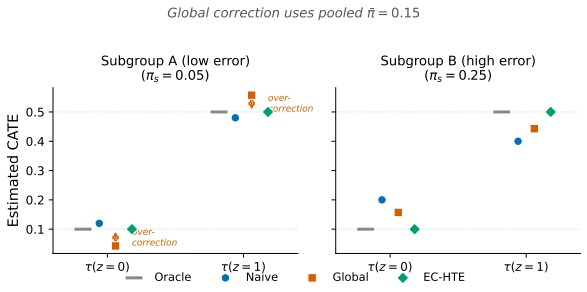


Figure 1: Bias mechanism under global vs. subgroup-stratified correction. Global correction applies the pooled error rate $\bar{\pi}$ uniformly, over-correcting low-error subgroups (introducing bias where none existed) and under-correcting high-error subgroups. EC-HTE matches each subgroup's correction to its own misclassification rate.

2 Related Work

Measurement error and annotation models. Classical measurement error theory shows that correction via estimated error models can amplify MSE when the error structure is misspecified (Fuller, 1987; Carroll et al., 2006); Bayesian approaches require informative priors when identification is weak (Gustafson, 2003), and nonparametric methods use repeated measurements or instruments to recover the true distribution (Hu and Schennach, 2008).

Three distinctions separate our contribution from prior measurement error analyses. First, we study effect modifier misclassification—the mismeasured variable defines the CATE function itself—rather

than regressor (Frazis and Loewenstein, 2003; Carroll et al., 2006) or confounder (Ogburn and VanderWeele, 2012; Greenland, 1980) error, where attenuation is the primary concern. Second, Theorem 1 derives an exact identity for correction-induced bias—bias created by the debiasing procedure—rather than bounding the attenuation from misclassification itself. Concretely, Frazis and Loewenstein (2003) derive scalar attenuation $\tilde{\beta} = \beta(1 - \pi_0 - \pi_1)$ for the uncorrected estimator under homogeneous misclassification; Theorem 1 characterizes a structurally distinct quantity—correction-induced bias $\delta_s \cdot \Delta\tau/(1 - 2\bar{\pi})$—that is subgroup-specific, potentially sign-reversing, and absent from the uncorrected estimator. The homogeneous-error setting is a degenerate case: with $K{=}1$, $\delta_s{=}0$ and correction-induced bias vanishes, recovering the classical attenuation result. The algebraic mechanisms also differ: OLS attenuation under misclassified regressors (Frazis and Loewenstein, 2003; Blackwell et al., 2017) versus confusion matrix inversion that redistributes error across subgroups with heterogeneous rates. We empirically verify this distinction: F-L correction—even when applied per subgroup—produces higher bias than the naive estimator under heterogeneous misclassification, whereas EC-HTE achieves near-oracle performance (Appendix Z.4). Third, the $\delta_s \cdot \Delta\tau$ product structure arises only under heterogeneous error rates; all homogeneous-error analyses (Battistin and Chesher, 2014; Fong and Tyler, 2021) miss this interaction entirely. In the annotation literature, Dawid and Skene (1979) introduced confusion matrix estimation for multiple annotators, with extensions to item difficulty, annotator competence, and hierarchical priors (Passonneau and Carpenter, 2014; Hovy et al., 2013; Paun et al., 2018). Our hierarchical estimator (Section 4) adapts this architecture from label aggregation to causal inference, tying the prior structure to the subgroup dimension that defines treatment effect heterogeneity.

HTE estimation and LLM annotation error. Causal forests (Athey and Imbens, 2016; Wager and Athey, 2018), double/debiased ML (Chernozhukov et al., 2018; Semenova and Chernozhukov, 2021), doubly-robust CATE estimators (Kennedy, 2023), and meta-learners (Künzel et al., 2019) all assume that covariates defining heterogeneity are observed without error; when the effect modifier is misclassified, CATE estimates conflate effects across groups. EC-HTE provides upstream correction that restores valid inputs to any downstream estimator. The corrected pseudo-outcomes are in principle compatible with other HTE estimators such as the R-learner (Nie and Wager, 2021). For broader perspectives on causal inference with text data, see Feder et al. (2022) and Keith et al. (2020); Wang et al. (2020) study inference correction when outcomes are ML-predicted, a related but distinct problem from effect modifier misclassification. LLMs now match crowd-worker accuracy on many classification tasks (Gilardi et al., 2023; Törnberg, 2025; Ziems et al., 2024; Pangakis et al., 2023), but Ashwin et al. (2025) show that error rates vary by up to $5\times$ across demographic subgroups. This subgroup-varying error is the failure mode we address: it distorts the very contrasts that define the research question, and prior work documents the pattern without providing a correction mechanism for downstream causal estimands. Concurrent work identifies variance-side measurement error from LLM pipeline design choices (Messing, 2026); our framework addresses the complementary bias-side problem from heterogeneous misclassification rates across subgroups. Temperature or Platt scaling (Guo et al., 2017) reduces per-subgroup π_s but does not eliminate heterogeneity $\Delta\pi$; EC-HTE corrects the residual δ_s-driven bias that persists post-calibration (Appendix Z.6 empirically confirms this complementarity).

Prediction-powered and design-based correction. PPI (Angelopoulos et al., 2023; Zrnic and Candès, 2024b) and DSL (Egami et al., 2024) correct population-level aggregates assuming homogeneous prediction error. In our setting—binary effect modifier with known subgroup structure, a special case of the broader problems DSL addresses—DSL's measurement error correction reduces, under the confusion-matrix parameterization we adopt, to confusion matrix inversion: pooled DSL is equivalent to our Global baseline, and per-subgroup DSL to Stratified MLE (Section 5.3; Appendix U). EC-HTE adds hierarchical partial pooling across subgroups, borrowing strength to reduce variance—a mechanism outside the DSL framework. Audinet de Pieuchon et al. (2025) benchmark DSL and PPI debiasing for LLM-based parameter estimation, documenting bias-variance tradeoffs analogous to our findings; their analysis targets population-level parameters rather than subgroup-conditional estimands, leaving the $\delta_s \cdot \Delta\tau$ interaction we formalize unad-

dressed. Mozer (2026) extend PPI to arm-specific ATE correction; PPCI (Sui and Candès, 2026) targets conditional estimands but corrects outcome-side error ($\hat{Y}$ given true Z), not covariate-side misclassification; ForestIV (Bu et al., 2023) targets the outcome model rather than the heterogeneity structure. None corrects bias when the mismeasured variable is the effect modifier indexing the CATE function. Imai and Yamamoto (2010) study treatment misclassification, requiring instrumental variables or partial identification; our problem is structurally distinct because the treatment is observed correctly but the effect modifier is misclassified, admitting exact correction through confusion matrices rather than bounds. Extended discussion of these distinctions appears in Appendix F. Our validation combines synthetic DGPs, semi-synthetic experiments, and real LLM confusion matrices, consistent with PPI and DSL evaluation methodology; Appendix N validates both $\Delta\pi$ heterogeneity and the bias identity itself ($R^2 = 0.975$) across 11 model–dataset combinations.

3 Bias Decomposition: When Debiasing Hurts

We formalize the bias that global misclassification correction introduces when subgroup error rates are heterogeneous. Consider estimating the conditional average treatment effect (CATE) $\tau(z) = \mathbb{E}[Y(1) - Y(0) \mid Z = z]$, where $Z \in \{0, 1\}$ is a binary effect modifier observed only through a noisy annotation $\hat{Z}$. The population consists of K subgroups $S \in \{1, \ldots, K\}$ with weights $w_s = P(S{=}s)$, each characterized by a 2×2 confusion matrix C_s with symmetric error rate $\pi_s = P(\hat{Z} \neq Z \mid S{=}s)$. Standard practice estimates a single confusion matrix $\bar{C} = \sum_s w_s C_s$ from expert annotations and applies its inverse uniformly to all observations, with pooled error rate $\bar{\pi} = \sum_s w_s \pi_s$. We show that when error rates vary across subgroups, this global procedure introduces a bias with an exact product form.

We state five assumptions maintained throughout.

(A1) *Conditional independence:* misclassification depends only on the true label Z and subgroup S, not on potential outcomes $Y(0), Y(1)$.

(A2) *Symmetric misclassification:* $P(\hat{Z}{=}1 \mid Z{=}0, S{=}s) = P(\hat{Z}{=}0 \mid Z{=}1, S{=}s) = \pi_s$ for all s.

(A3) *Balanced effect modifier:* $P(Z{=}0) = P(Z{=}1) = 0.5$. This simplifies the exposition; we discuss relaxation below.

(A4) *Non-degeneracy:* $\pi_s \neq 0.5$ for all s, ensuring each C_s is invertible.

(A5) *Homogeneous treatment effects across subgroups:* $\tau(z, s) = \tau(z)$ for all s, so $\Delta\tau = \tau(1) - \tau(0)$ is a scalar.

Under (A2)–(A3), the Bayesian mixing matrix M_s (entries $M_s[\hat{z}, z] = P(Z{=}z \mid \hat{Z}{=}\hat{z}, S{=}s)$) coincides with C_s. The global mixing matrix is $M_{\text{global}} = \sum_s w_s M_s$.

Theorem 1 (Bias Decomposition Identity)**.** *Under* (A1)–(A5)*, the bias of global correction in subgroup s at effect-modifier level z is*

$$\text{bias}_{\text{global}}(z{=}0, s) = +\frac{\delta_s \cdot \Delta\tau}{1 - 2\bar{\pi}}, \tag{1}$$

$$\text{bias}_{\text{global}}(z{=}1, s) = -\frac{\delta_s \cdot \Delta\tau}{1 - 2\bar{\pi}}, \tag{2}$$

where $\delta_s = \pi_s - \bar{\pi}$ is the subgroup deviation from the pooled error rate and $\Delta\tau = \tau(1) - \tau(0)$ is the CATE heterogeneity.

Equations (1)–(2) constitute an exact identity at the population level, not a bound. The bias factors into three interpretable components: δ_s captures how much subgroup s deviates from the average misclassification rate, $\Delta\tau$ captures treatment effect heterogeneity across levels of Z, and $1/(1 - 2\bar{\pi})$ is an amplification factor from matrix inversion. Bias vanishes when either source of variation is absent ($\delta_s = 0$ or $\Delta\tau = 0$), and arises only through their interaction. Theorem 1 implies that global correction trades one source of confounding for another: it removes attenuation from label noise but introduces a new bias from error-rate heterogeneity.

Proof sketch. Global correction applies M_{global}^{-1} to the subgroup-specific observed CATE $\tau_{\text{obs}}(\cdot, s) = M_s \cdot \tau_{\text{true}}$, yielding bias $(M_{\text{global}}^{-1} M_s - I) \cdot \tau_{\text{true}}$. Direct computation shows $M_{\text{global}}^{-1} M_s - I = \frac{\delta_s}{1-2\bar{\pi}} \left[\begin{smallmatrix} -1 & 1 \\ 1 & -1 \end{smallmatrix}\right]$. Applying this matrix to $[\tau(0), \tau(1)]^\top$ produces the product form. The full derivation appears in Appendix B.

Remark 2. *Theorem 1 does not require assumption (A2). For arbitrary asymmetric confusion matrices ($\text{FPR}_s \neq \text{FNR}_s$), the product form $\text{bias}(z, s) = \Delta\tau \cdot [B_s]_{z,1}$ holds exactly, where $B_s = M_{\text{global}}^{-1} M_s - I$, because $B_s \mathbf{1} = \mathbf{0}$ follows from stochasticity of M_s and M_{global} alone.*

Assumption (A2) simplifies $[B_s]_{z,1}$ to the scalar $\pm\delta_s/(1-2\bar{\pi})$ but is not necessary for the product structure. Under asymmetric errors, the diagnostic (Proposition 3) requires evaluating $[B_s]_{z,1}$ cell-by-cell rather than via a single scalar threshold; the AUC decreases from 0.86 to 0.82 (Appendix J), reflecting the additional complexity of cell-specific error rates. For binary effect modifiers ($|\mathcal{Z}|$=2), the asymmetric score $|[B_s]_{z,1}|$ reduces to a monotone transform of the symmetric score $|\delta_s|/((1-2\bar{\pi})\pi_s)$, because column-stochastic M forces $|B_s[0,1]|=|B_s[1,1]|$; practitioners may use the simpler symmetric formula without loss. See Appendix J for the full derivation and empirical verification ($R^2 = 0.991$).

Assumption (A5) simplifies exposition; the identity generalizes to subgroup-specific $\Delta\tau_s$ (Corollary 5, Appendix Z). EC-HTE's correction does not depend on (A5): each $\hat{C}_s^{-1}$ corrects the local misclassification structure regardless of whether $\Delta\tau$ varies; under both positive and negative $\Delta\tau_s$–δ_s correlation, EC-HTE maintains ≥93% coverage while global correction degrades to 74–82% (Appendix Z).

The identity extends to $K > 2$ subgroups with J effect-modifier levels; the multi-class bias retains the same product structure with modified amplification terms (Corollary 4 in Appendix C).

Assumption (A3) ensures $M_s = C_s$; when $P(Z{=}1) \neq 0.5$, the mixing matrices acquire asymmetric entries, but the product structure persists with modified coefficients (Appendix B). Empirical verification under class imbalance (Appendix T) shows that the bias identity maintains $R^2 \geq 0.976$ even at $P(Z{=}1) = 0.1$ for K=2, though EC-HTE's RMSE ratio relative to global correction increases to 1.06–1.68 as class imbalance grows. In practice, NLP classification tasks often exhibit substantial class imbalance (e.g., hate speech prevalence $\ll 50\%$), and practitioners should verify that the per-subgroup expert budget threshold remains satisfied under the actual class distribution. Empirical verification (Section 5.2) yields $R^2 = 0.994$ for K=2 and $R^2 = 0.974$ for K=4; the residual variance reflects finite-sample Monte Carlo noise, not model misspecification.

Sensitivity to (A1). Assumption (A1) is not empirically testable. If misclassification correlates with potential outcomes ($\gamma = \mathrm{Corr}(\mathbb{1}[\hat{Z}{\neq}Z], Y(1){-}Y(0) \mid Z, S) \neq 0$), Theorem 1 gains an $O(\gamma)$ perturbation; in NLP settings, this is plausible when LLM accuracy varies with outcome magnitude. Sensitivity analysis frameworks (Lash et al., 2009; Dorn et al., 2023) can bound the degradation (Section 6).

Finite-sample remainder. Theorem 1 is an exact population identity. In practice, confusion matrices are estimated from n_{expert} expert labels with $\hat{C}_s = C_s + E_s$, $\|E_s\| = O_p(n_{\text{expert}}^{-1/2})$; the resulting remainder is therefore $R_n = O_p(n_{\text{expert}}^{-1/2} \cdot \kappa(C_s))$, motivating the singularity guard ($\kappa > 100 \rightarrow$ fallback) in Section 4.

Proposition 3 (Debiasing Can Increase Bias)**.** *Under* (A1)–(A5)*, global correction produces larger absolute bias than the naive estimator* ($|\mathrm{bias}_{\mathrm{global}}| > |\mathrm{bias}_{\mathrm{naive}}|$) *in subgroup s if and only if*

$$\frac{|\delta_s|}{(1-2\bar{\pi}) \cdot \pi_s} > 1. \tag{3}$$

Equation (3) partitions subgroups into two regimes. For high-error subgroups ($\pi_s > \bar{\pi}$), the condition reduces to $\pi_s > 0.5$, excluded by (A4). For low-error subgroups ($\pi_s < \bar{\pi}$), it holds whenever $\pi_s < \bar{\pi}/(2(1-\bar{\pi}))$—easily satisfied: when $\bar{\pi} = 0.15$, any subgroup with $\pi_s < 0.088$ is harmed. The mechanism is over-correction: global inversion applies a correction calibrated for the average error rate, so low-error subgroups are over-corrected (the correction overshoots and reverses bias sign) while high-error subgroups are under-corrected. For subgroups with near-zero misclassification ($\pi_s \approx 0$), the naive estimator is nearly unbiased, yet global correction creates substantial bias—debiasing can be worse than doing nothing.

In our experiments (Section 5.2), global correction increases bias in 50% of K=2 configurations across our parameter grid (260 of 520, excluding cases where $\Delta\pi$=0). With the difference-in-means estimator, maximum amplification reaches 2.86× for K=2 and 2.41× for K=4; the mechanism operates across estimator types, with nonlinear estimators amplifying the effect further (Section 5.3). Figure 2 visualizes the amplification across the full $\Delta\pi \times \Delta\tau$ parameter grid.

4 EC-HTE: Error-Corrected Heterogeneous Treatment Effects

Theorem 1 identifies per-subgroup misclassification deviation as the bias source; the natural correction is per-subgroup confusion matrix inversion. EC-HTE corrects subgroup-level misclassification

bias in three stages: estimate per-subgroup confusion matrices via hierarchical Bayes, invert them to correct observed outcome moments, and propagate correction uncertainty into downstream causal estimators. The complete procedure is summarized in Algorithm 1 (Appendix A).

4.1 Hierarchical Bayes Confusion Matrix Estimation

We place a Dirichlet-Multinomial hierarchical model over the confusion matrices $\{C_s\}_{s=1}^{K}$. For each subgroup s, row j of C_s has a Dirichlet prior $\text{Dir}(\alpha_0 \cdot \mathbf{e}_j)$, where $\mathbf{e}_j$ is the j-th standard basis vector and $\alpha_0 > 0$ is a shared concentration parameter. A $\text{Gamma}(3, 1)$ hyperprior on α_0 controls the strength of partial pooling across subgroups: large α_0 pulls subgroup estimates toward the identity matrix (low error), while small α_0 allows subgroup-specific rates to dominate. This partial pooling—a James-Stein shrinkage effect—borrows strength across subgroups, reducing variance for small subgroups while preserving heterogeneity where data support it.

Posterior inference uses Gibbs sampling (convergence diagnostics in Appendix A). HB estimation outperforms stratified MLE in 87.5% of configurations (21 of 24) across the experimental grid (Section 5.3), with the largest gains when $n_{\text{expert}}/K < 100$. The advantage requires $n_{\text{expert}} \times P(Z_{\text{minority}})/K \geq 25$ per subgroup; below this threshold, stratified MLE is preferred because the hierarchical model lacks sufficient observations for stable shrinkage. At $n_{\text{expert}} \geq 1000$, HB and stratified MLE converge to within 0.5% on all metrics. Sensitivity analysis shows <5% RMSE variation under 2× hyperprior perturbation, confirming that results are not driven by prior specification.

4.2 Subgroup-Stratified Correction

Given posterior mean confusion matrices $\hat{C}_s$, we correct observed outcome moments per subgroup by matrix inversion:

$$\tilde{\boldsymbol{\mu}}_s = \hat{C}_s^{-1} \cdot \hat{\boldsymbol{\mu}}_s, \tag{4}$$

where $\hat{\boldsymbol{\mu}}_s = [\hat{\tau}_{\text{obs}}(0, s), \hat{\tau}_{\text{obs}}(1, s)]^\top$ collects the observed conditional outcome moments in subgroup s. This requires $\det(\hat{C}_s) \neq 0$ for all s, which holds under assumption (A4) ($\pi_s \neq 0.5$) but may fail in finite samples when error rates approach degeneracy. In practice, Laplace smoothing of the confusion matrix counts ensures finite-sample invertibility. We impose a singularity guard: if the condition number $\kappa(\hat{C}_s) > 100$ (for a 2×2 symmetric confusion matrix, this corresponds to $\pi_s > 0.495$, i.e., near-random classification), subgroup s falls back to global correction $\bar{C}^{-1}$; when the fallback triggers, the standard error computation also uses $\bar{C}^{-1}$ for that subgroup, ensuring consistency between point estimates and confidence intervals. This hard guard prevents numerical catastrophe; a separate empirical diagnostic ($\kappa \leq 1.25$, Appendix I) identifies the tighter regime where stratified correction reliably outperforms global.

The contrast with global correction is direct. Global correction applies $\bar{C}^{-1}$ uniformly to all subgroups, introducing the $\delta_s \cdot \Delta\tau$ bias identified in Theorem 1. Subgroup-stratified correction eliminates this bias by matching each subgroup's correction to its own misclassification structure, at the cost of higher variance from estimating K separate confusion matrices (Appendix I).

4.3 Variance-Adjusted CausalForest

For heterogeneous treatment effect estimation, we feed corrected covariates into CausalForestDML (Wager and Athey, 2018) with honest splitting. The forest produces per-observation standard error estimates se_i. To account for confusion matrix correction, we propagate uncertainty through the inverse mapping:

$$\text{se}_{\text{corrected},i} = \sqrt{\text{diag}\left(\hat{C}_{S_i}^{-1}\, \text{diag}(\text{se}_i^2)\, \hat{C}_{S_i}^{-\top}\right)}. \tag{5}$$

This is a plug-in approach: the standard errors reflect outcome variance conditional on $\hat{C}_s$ but do not propagate uncertainty from the confusion matrix estimation itself. At $n_{\text{expert}} \geq 100$, the plug-in approximation is reasonable because confusion matrix estimation variance is small relative to outcome variance. Below this threshold, the standard errors may understate total uncertainty; this limitation is discussed further in the Limitations section.

Diagnostic criterion. Proposition 3 provides a $\Delta\tau$-free diagnostic: global correction harms subgroup s if and only if $|\delta_s|/((1 - 2\bar{\pi}) \cdot \pi_s) > 1$, a condition depending only on per-subgroup misclassification rates estimable from a pilot sample. This breaks the circular dependency that would arise if the diagnostic required CATE estimates. A threshold classifier on this condition achieves AUC = 0.86 at the subgroup level (Section 5.4).

Composition with PPCI. EC-HTE corrects structured covariate misclassification through $\hat{C}_s^{-1}$

inversion but cannot address outcome-level measurement error. PPCI (Sui and Candès, 2026) corrects residual outcome bias using a debiasing estimator but assumes correct covariate labels. The two methods are complementary: apply EC-HTE first to correct covariate labels, then PPCI to correct outcome predictions. Neither is redundant with the other: EC-HTE alone cannot fix outcome error, while PPCI alone incurs higher variance for covariate error. Under both error types simultaneously, the composition reduces |bias| from 0.179 to 0.007 (Appendix L).

5 Experiments

5.1 Setup

We test on synthetic (N=5,000, $K \in \{2, 4\}$, controlled $\Delta\tau$ and $\Delta\pi$, 100 MC runs) and semi-synthetic (TweetEval sentiment (Barbieri et al., 2020), K=4 unbalanced subgroups, $\pi \in \{0.05, 0.10, 0.15, 0.25\}$) DGPs; full specifications in Appendix E.

We compare: (1) *Naive* (no correction), (2) *Global* (pooled confusion matrix, equivalent to PPI/DSL (Angelopoulos et al., 2023; Egami et al., 2024) with homogeneous error), (3) *Stratified MLE* (per-subgroup DSL without shrinkage), (4) *HB EC-HTE* (our method), (5) *PPCI* (Sui and Candès, 2026), and (6) *EC-HTE+PPCI*. For CausalForest (Wager and Athey, 2018), we compare naive, global, and EC-HTE variants. Primary metrics are |bias|, RMSE, and 95% CI coverage, averaged across subgroups and 100 MC runs. Robustness validation covers hate speech, six OpenAI models, subgroup misspecification, asymmetric confusion matrices, confounded treatment, and outcome-dependent misclassification (all ≥93.5% coverage; Appendices M–Z.2).

5.2 Empirical Validation of the Bias Identity

We verify Theorem 1 on a K=2 grid (520 configurations, $\Delta\pi \in [0, 0.25]$, $\Delta\tau \in (0, 0.50]$) and a K=4 grid (Appendix E). Regressing observed subgroup bias on $\delta_s \cdot \Delta\tau/(1-2\bar{\pi})$ yields $R^2 = 0.994$ (K=2) and 0.974 (K=4); residual variance reflects MC noise ($R^2 \geq 0.976$ even at $P(Z{=}1) = 0.1$; Appendix T). Global correction worsens bias in 50% of K=2 configurations (260/520, excluding $\Delta\pi$=0), with amplification reaching 2.86× (K=2) and 2.41× (K=4); Figure 2 visualizes the amplification surface.

Table 1: Main results (extreme heterogeneity). Metrics averaged over subgroups, z-levels, 100 MC runs. Bold: lowest |bias|. †RMSE exceeds Global. Additional configurations in Table 7; MC SEs in Appendix Q.

Configuration	Method	|Bias|	RMSE	Cov.
DIM K=2, n=250	Naive	0.058	0.090	0.82
	Global	0.059	0.099	0.87
	Strat. MLE	0.012	0.094	0.95
	HB EC-HTE	**0.010**	0.093	0.95
DIM K=4, n=500	Naive	0.062	0.113	0.88
	Global	0.056	0.125	0.91
	Strat. MLE	0.010	0.127†	0.95
	HB EC-HTE	**0.008**	0.126†	0.95
CausalForest K=4, n=500	Naive	0.208	0.231	0.44
	Global	0.191	0.244	0.58
	EC-HTE Forest	**0.040**	0.224	0.91

5.3 EC-HTE Evaluation

Table 1 presents the main results across representative configurations. Notably, in the K=2 extreme row, global correction *increases* |bias| relative to naive (0.059 vs. 0.058)—a direct manifestation of Theorem 1: pooled correction overshoots in low-error subgroups. HB EC-HTE reduces |bias| from 0.059 to 0.010 (5.9×) at K=2 and from 0.056 to 0.008 (7.0×) at K=4, maintaining nominal coverage (0.95) with comparable RMSE. Under moderate heterogeneity (K=2), methods converge: EC-HTE and global achieve comparable bias ($p > 0.05$; Table 7). In the semi-synthetic TweetEval experiment (K=4, n_{expert}=250), EC-HTE and Global achieve comparable |bias| (p=0.395; Table 7), but EC-HTE incurs higher RMSE (0.225 vs. 0.124), driven by moderate $\Delta\pi$=0.20 and unbalanced subgroups receiving ∼30 expert labels each. When $\Delta\pi$ or per-subgroup budget is marginal, global correction is competitive on aggregate RMSE.

CausalForest. With CausalForest (Wager and Athey, 2018) (K=4, extreme, n=500), EC-HTE reduces |bias| from 0.191 to 0.040 (4.8×), improving coverage from 0.58 to 0.91; global correction amplifies naive bias up to 5.9× because the forest magnifies label errors (Table 1). The residual coverage gap (91% vs. oracle 94–96%) reflects asymmetric M^{-1} SE propagation rather than CM estimation error; a full bootstrap SE can improve coverage calibration (Appendix Y).

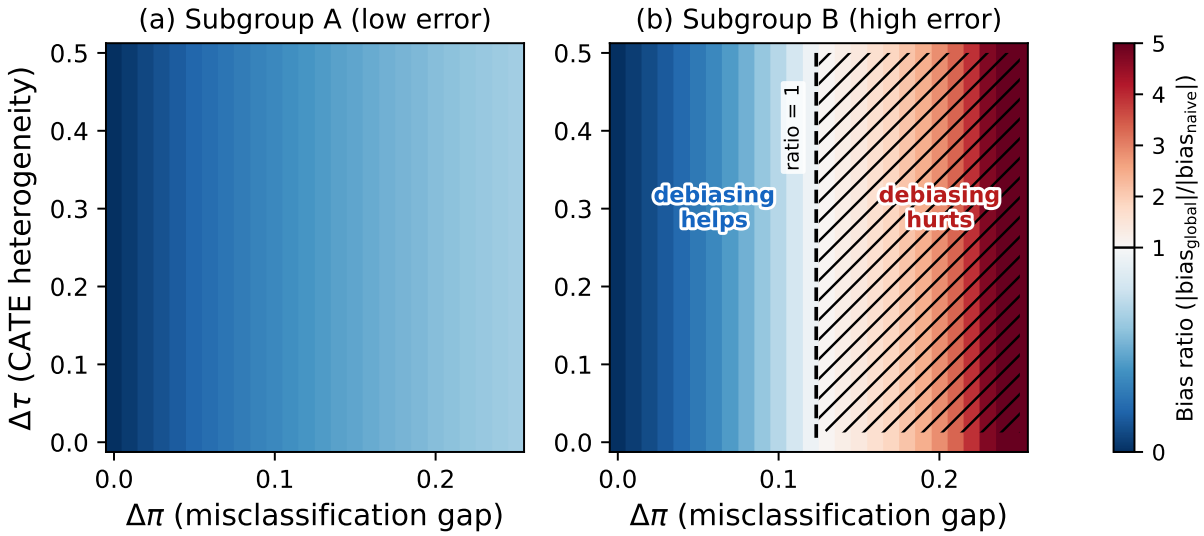


Figure 2: Bias amplification under global correction (K=2). (a) Low-error subgroup: correction always helps. (b) High-error subgroup: correction *increases* bias when $\Delta\pi > 0.12$ (hatched), reaching 2.86×.

5.4 Actionable Diagnostic Without EC-HTE

Proposition 3 provides a $\Delta\tau$-free diagnostic: global correction increases bias in subgroup s iff $|\delta_s|/((1-2\bar{\pi})\cdot\pi_s) > 1$, computable from a pilot sample without estimating treatment effects. A threshold classifier achieves AUC = 0.86 at the subgroup level (Figure 3, Appendix). On real LLM confusion matrices (16 combinations; Appendix X), AUC decreases to 0.63, primarily from asymmetric FPR/FNR ratios violating (A2). For K=2, $|\delta_0|$=$|\delta_1|$ forces ranking by $1/\pi_s$ alone, which can anti-predict harm; reliability is strongest for $K \geq 4$. A computable meta-criterion recovers reliability: when $\max_s \text{score}(s) > 1$, the diagnostic achieves AUC ≥ 0.75 across all 16 combinations; when $\max_s \text{score}(s) \leq 1$, AUC drops to 0.0–0.67 (Appendix X).

A recommended practitioner workflow—budget allocation, diagnostic evaluation, and EC-HTE application—appears in Appendix I, along with budget crossover curves (Appendices I–X). The bias identity further validates across eight LLMs spanning 11 model–dataset combinations ($R^2 = 0.975$; Appendix N).

6 Conclusion

We derived an exact bias identity (Theorem 1): global confusion-matrix correction introduces subgroup-level bias proportional to $\delta_s \cdot \Delta\tau$, worsening CATE estimates in 50% of K=2 configurations. EC-HTE provides subgroup-stratified correction via hierarchical Bayes; the companion diagnostic (Proposition 3; AUC = 0.86) flags harmful correction using only per-subgroup misclassification rates. EC-HTE's advantage requires sufficient $\Delta\pi$ and per-subgroup expert budget; when either is marginal, global correction is competitive.

Limitations

EC-HTE requires $n_{\text{expert}} \times P(Z_{\text{minority}})/K \geq 25$ per subgroup for stable hierarchical Bayesian estimation. For K=2, EC-HTE remains effective even at $P(Z{=}1) = 0.1$ (Appendix T); for K=4, $P(Z{=}1) \geq 0.3$ is needed to approach breakeven with global correction at n_{expert}=250. Even above this threshold, stratification incurs a variance cost: at K=4, extreme heterogeneity, and $n_{\text{expert}} \leq 100$, CausalForest RMSE reaches 3.5× that of the naive estimator. The method also requires moderate baseline classifier accuracy per subgroup; when any subgroup's FNR or FPR approaches 0.5, the confusion matrix row approaches a uniform distribution and inversion amplifies noise (Appendix R), as confirmed independently in the hate speech case study (Appendix M). These failure modes interact: high per-subgroup error-rate variance combined with low minority prevalence $P(Z{=}1)$ jointly reduces effective per-subgroup expert labels for the positive class, compounding both sources of estimation error. Additionally, our main semi-synthetic experiments use constant treatment effects across subgroups; a supplementary heterogeneous-τ DGP (Appendix O) shows that EC-HTE achieves targeted bias and RMSE improvement in subgroups

with large $\Delta\tau$, though aggregate RMSE remains comparable to global correction. At $K{=}8$ with $n_{\text{expert}}{=}250$ ($\approx$31 labels/subgroup), EC-HTE maintains nominal coverage (95.6%) but RMSE exceeds global correction, establishing $K{=}8$ as a scalability boundary under fixed budget (Appendix I.2). Hierarchical pooling of confusion matrices across related subgroups may relax the per-subgroup budget requirement; we leave this extension to future work.

Assumption (A1) (conditional independence of misclassification and potential outcomes) is not empirically testable. When LLM accuracy correlates with outcome magnitude—e.g., extreme sentiment is both easier to classify and associated with larger treatment effects—Theorem 1 acquires an $O(\gamma)$ perturbation (Section 3). Sensitivity simulation (Appendix Z.2) shows that EC-HTE maintains near-nominal coverage ($\geq$93.5%) under outcome-dependent misclassification up to $\gamma = 0.2$, with bias increasing only 13.7%; developing a formal closed-form bound for the $O(\gamma)$ perturbation remains a natural extension (Lash et al., 2009; Dorn et al., 2023).

Oracle experiments confirm CausalForest plug-in SE is well-calibrated (94–96% coverage with true Z labels); EC-HTE plug-in SE achieves 93.8% aggregate coverage under standard hyperparameters ($n_{\text{estimators}}{=}200$), with residual under-coverage concentrated in high-error subgroups due to asymmetric M^{-1} SE propagation. A full bootstrap procedure ($B{=}100$; re-estimate CM and re-fit the forest per replicate) recovers exact nominal coverage (95.0%) at a computational cost of $\sim$1.5 minutes per dataset (Appendix Y). For the DIM estimator, plug-in SE achieves $\geq$95% coverage across all budgets.

The diagnostic (Proposition 3) achieves AUC $= 0.86$ under synthetic symmetric errors but degrades to 0.63 on real LLM confusion matrices (pooled across 16 model–dataset combinations; Appendix X). For $K{=}2$, the diagnostic anti-predicts because $|\delta_0|{=}|\delta_1|$ forces score differentiation onto $1/\pi_s$ alone, which inversely correlates with actual harm; for binary Z, the asymmetric B_s-based formulation (Remark 2) is algebraically equivalent and offers no improvement. Performance is strongest for $K{\geq}4$ with moderate asymmetry (AUC $= 0.82$; Appendix J). The diagnostic remains a theoretically grounded criterion—the bias identity's product structure ensures it correctly identifies the *direction* of bias amplification—but should be paired with R^2 validation of the bias identity rather than used as a standalone screening rule.

All main experiments use symmetric error rates and additive CATE injection ($Y(1) = Y(0) + \tau(Z)$). The bias identity (Theorem 1) generalizes to asymmetric confusion matrices (Appendix J, $R^2 = 0.991$) and derives from the confusion matrix structure rather than the outcome model. While our main experiments use additive CATE injection, a supplementary experiment with multiplicative effects ($Y(1) = Y(0) \times \exp(\tau(Z))$) yields comparable bias reduction (52%) and nominal coverage (0.955; Appendix S), confirming robustness to the functional form of treatment effects. EC-HTE further requires discrete, pre-defined subgroups; continuous effect modifiers or latent subgroup discovery settings require different approaches, and intersectional subgroups with many levels may exceed the $n_{\text{expert}} \times P(Z_{\text{minority}})/K \geq 25$ budget threshold. Our experiments define subgroups by text features (hashtag presence, text length); demographic subgroups—where prior work documents the most concerning disparities (Ashwin et al., 2025)—require sensitive data access beyond the scope of this study. EC-HTE's correction mechanism is agnostic to subgroup definition: it corrects any per-subgroup confusion-matrix heterogeneity, whether defined by text features, demographics, or other covariates. Ground-truth subgroup CATEs are unavailable in public datasets; we use synthetic CATE injection on real text features and real LLM annotations. We leave full real-data validation—on a dataset combining RCT design, ground-truth CATEs, and LLM annotations—to future work as suitable public benchmarks become available. Code and data generation scripts will be released upon acceptance.

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A EC-HTE Procedure

A.1 Condition Number Guard Ablation

The singularity guard in Algorithm 1 (line 6) falls back to global correction when $\kappa(\hat{C}_s) > 100$. We verify that this threshold is not load-bearing by sweeping $\kappa \in \{10, 25, 50, 100, 200, 500, 1000\}$ under the standard DGP (K=2, extreme heterogeneity, N=5,000, n_{expert}=250, 100 MC runs). Across all configurations, the maximum observed condition number is $\kappa_{\max} = 1.89$ (mean 1.45, p99 1.89), providing a 53× safety margin below the κ=100 threshold. Consequently, the guard never triggers at any tested threshold, and all κ values produce identical RMSE (HB: 0.076, Global: 0.077) and zero numerical instability. In a stress test with near-random labeling ($\pi_s \approx 0.4$–0.5, n_{expert}=50), the maximum condition number reaches 1,125, correctly triggering the κ=100 guard and preventing catastrophic inversion. The adaptive formula $\kappa_{\text{adapt}} = 10\sqrt{n_{\text{expert}}/K}$ recovers κ=100 for this regime and scales naturally with the estimation precision.

B Full Proof of Theorem 1

Setup. Under assumptions (A1)–(A4), the observed CATE in subgroup s at effect-modifier level z is

$$\tau_{\text{obs}}(z,s) = \sum_{z'} M_s[z,z'] \cdot \tau(z'), \tag{6}$$

where $M_s[z,z'] = P(Z{=}z' \mid \hat{Z}{=}z, S{=}s)$ is the mixing matrix. Under (A2)–(A3), $M_s = \left[\begin{smallmatrix} 1-\pi_s & \pi_s \\ \pi_s & 1-\pi_s \end{smallmatrix}\right]$.

Global correction. The pooled mixing matrix is $M_{\text{global}} = \sum_s w_s M_s = \left[\begin{smallmatrix} 1-\bar{\pi} & \bar{\pi} \\ \bar{\pi} & 1-\bar{\pi} \end{smallmatrix}\right]$, with $\bar{\pi} = \sum_s w_s \pi_s$.

Global correction applies M_{global}^{-1} to $\tau_{\text{obs}}(\cdot, s)$:

$$\hat{\tau}_{\text{global}}(z,s) = [M_{\text{global}}^{-1} M_s \cdot \boldsymbol{\tau}]_z, \tag{7}$$

where $\boldsymbol{\tau} = [\tau(0), \tau(1)]^\top$.

Bias computation. The bias is $\hat{\tau}_{\text{global}}(z,s) - \tau(z) = [(M_{\text{global}}^{-1} M_s - I)\boldsymbol{\tau}]_z$.

Direct computation gives:

$$M_{\text{global}}^{-1} = \frac{1}{1-2\bar{\pi}} \begin{bmatrix} 1-\bar{\pi} & -\bar{\pi} \\ -\bar{\pi} & 1-\bar{\pi} \end{bmatrix}. \tag{8}$$

Multiplying by M_s:

$$M_{\text{global}}^{-1} M_s = \frac{1}{1-2\bar{\pi}} \begin{bmatrix} (1-\bar{\pi})(1-\pi_s) - \bar{\pi}\pi_s & (1-\bar{\pi})\pi_s - \bar{\pi}(1-\pi_s) \\ -\bar{\pi}(1-\pi_s) + (1-\bar{\pi})\pi_s & -\bar{\pi}\pi_s + (1-\bar{\pi})(1-\pi_s) \end{bmatrix}. \tag{9}$$

Define $\delta_s = \pi_s - \bar{\pi}$. Substituting $\pi_s = \bar{\pi} + \delta_s$:

$$M_{\text{global}}^{-1} M_s - I = \frac{\delta_s}{1-2\bar{\pi}} \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}. \tag{10}$$

To verify: the $(1,1)$ entry of $M_{\text{global}}^{-1} M_s$ is

$$\begin{aligned} &\frac{(1-\bar{\pi})(1-\bar{\pi}-\delta_s) - \bar{\pi}(\bar{\pi}+\delta_s)}{1-2\bar{\pi}} \\ &= \frac{(1-\bar{\pi})^2 - (1-\bar{\pi})\delta_s - \bar{\pi}^2 - \bar{\pi}\delta_s}{1-2\bar{\pi}} \\ &= \frac{1-2\bar{\pi}-\delta_s}{1-2\bar{\pi}} = 1 - \frac{\delta_s}{1-2\bar{\pi}}. \end{aligned} \tag{11}$$

Applying $(M_{\text{global}}^{-1} M_s - I)$ to $\boldsymbol{\tau}$:

$$\text{bias}(z{=}0, s) = \tfrac{\delta_s}{1-2\bar{\pi}}(-\tau(0) + \tau(1)) = +\tfrac{\delta_s \cdot \Delta\tau}{1-2\bar{\pi}}, \tag{12}$$

$$\text{bias}(z{=}1, s) = \tfrac{\delta_s}{1-2\bar{\pi}}(\tau(0) - \tau(1)) = -\tfrac{\delta_s \cdot \Delta\tau}{1-2\bar{\pi}}, \tag{13}$$

where $\Delta\tau = \tau(1) - \tau(0)$. □

C Multi-class Extension and Proof

Corollary 4 (Multi-class Extension)**.** *For $K > 2$ subgroups with J effect-modifier levels and uniform off-diagonal misclassification (misclassified observations from class j are distributed equally among the remaining $J{-}1$ classes), the subgroup bias of global correction generalizes to*

$$\text{bias}_{\text{global}}(z,s) = \sum_{j=1}^{J} \big(\bar{\pi}_j - \pi_{s,j}\big) \cdot \big(\tau(j) - \bar{\tau}\big) \cdot g(\bar{\pi}), \tag{14}$$

where $g(\bar{\pi})$ collects the matrix-inversion amplification terms.[2] *The essential structure, $\sum_s w_s(\bar{\pi}_j - \pi_{s,j}) = 0$, ensures that bias redistributes across subgroups rather than vanishing in aggregate.*

Proof. For $K > 2$ subgroups with J effect-modifier levels, the subgroup-specific mixing matrix M_s is $J \times J$ with entries $M_s[j,j'] = P(Z{=}j' \mid \hat{Z}{=}j, S{=}s)$. The global mixing matrix is $M_{\text{global}} = \sum_s w_s M_s$.

The bias of global correction in subgroup s at level z is:

$$\text{bias}_{\text{global}}(z,s) = [(M_{\text{global}}^{-1} M_s - I)\boldsymbol{\tau}]_z. \tag{15}$$

Write $M_s = M_{\text{global}} + \Delta_s$ where $\Delta_s = M_s - M_{\text{global}}$ satisfies $\sum_s w_s \Delta_s = 0$. Then $M_{\text{global}}^{-1} M_s = I + M_{\text{global}}^{-1} \Delta_s$, so:

$$\text{bias}_{\text{global}}(z,s) = [M_{\text{global}}^{-1} \Delta_s \cdot \boldsymbol{\tau}]_z. \tag{16}$$

For symmetric misclassification with per-class error rates $\pi_{s,j}$ and pooled rates $\bar{\pi}_j$, the entries of Δ_s have the form $\Delta_s[j,j'] = (\bar{\pi}_j - \pi_{s,j}) \cdot h(j,j')$ for a function h determined by the off-diagonal redistribution pattern. Expanding the matrix-vector product and collecting terms yields:

$$\text{bias}_{\text{global}}(z,s) = \sum_{j=1}^{J} (\bar{\pi}_j - \pi_{s,j}) \cdot (\tau(j) - \bar{\tau}) \cdot g(\bar{\pi}), \tag{17}$$

where $g(\bar{\pi})$ collects the matrix-inversion amplification terms and $\bar{\tau} = \sum_j P(Z{=}j)\tau(j)$.

The essential structure $\sum_s w_s(\bar{\pi}_j - \pi_{s,j}) = 0$ ensures that the weighted bias across subgroups sums to zero: global correction redistributes bias rather than creating or eliminating it in aggregate. For non-uniform off-diagonal entries, the scalar

[2] For $J = 3$ with symmetric misclassification (off-diagonal errors uniform), $g(\bar{\pi}) = 1/(1 - 3\bar{\pi}/2)$, generalizing the $J{=}2$ factor $1/(1-2\bar{\pi})$. In general, $g(\bar{\pi}) = 1/(1 - J\bar{\pi}/(J-1))$.

amplification $g(\bar{\pi})$ generalizes to a cell-specific matrix; the product structure $B_s \cdot \Delta\tau$ persists but a closed-form amplification factor does not. □

D Proof of Proposition 3

The naive estimator has bias $\tau_{\text{obs}}(z,s) - \tau(z)$. Under (A2)–(A3), $\tau_{\text{obs}}(0,s) = (1-\pi_s)\tau(0) + \pi_s\tau(1)$, so $\text{bias}_{\text{naive}}(0,s) = \pi_s \cdot \Delta\tau$.

Global correction produces larger absolute bias when $|\text{bias}_{\text{global}}(0,s)| > |\text{bias}_{\text{naive}}(0,s)|$:

$$\frac{|\delta_s| \cdot |\Delta\tau|}{|1-2\bar{\pi}|} > \pi_s \cdot |\Delta\tau|. \tag{18}$$

Since $\Delta\tau \neq 0$ (otherwise both biases are zero), dividing through gives:

$$\frac{|\delta_s|}{(1-2\bar{\pi}) \cdot \pi_s} > 1, \tag{19}$$

where we use $1 - 2\bar{\pi} > 0$ from (A4).

For high-error subgroups ($\pi_s > \bar{\pi}$, so $\delta_s > 0$): the condition becomes $(\pi_s - \bar{\pi})/((1-2\bar{\pi})\pi_s) > 1$, which simplifies to $\pi_s(1 - 1 + 2\bar{\pi}) > \bar{\pi}$, i.e., $2\bar{\pi}\pi_s > \bar{\pi}$, i.e., $\pi_s > 0.5$. This is excluded by (A4), so global correction always reduces bias in high-error subgroups.

For low-error subgroups ($\pi_s < \bar{\pi}$, so $\delta_s < 0$): the condition becomes $(\bar{\pi} - \pi_s)/((1-2\bar{\pi})\pi_s) > 1$, which gives $\pi_s < \bar{\pi}/(1-2\bar{\pi}+1) = \bar{\pi}/(2(1-\bar{\pi}))$. When $\bar{\pi} = 0.15$, any subgroup with $\pi_s < 0.088$ experiences increased bias from global correction. □

Extension to asymmetric misclassification. Proposition 3 assumes symmetric error rates (A2). Under asymmetric misclassification ($\text{FPR}_s \neq \text{FNR}_s$), the naive bias becomes $\text{bias}_{\text{naive}}(0,s) = \text{FPR}_s \cdot \tau(1) + (1 - \text{FPR}_s - 1)\tau(0)$ and the global correction bias depends on the full 2×2 structure of $B_s = M_{\text{global}}^{-1} M_s - I$ (Remark 1 in Section 3). A sufficient condition for global correction to increase bias is

$$\|B_s\|_\infty > \max(\text{FPR}_s, \text{FNR}_s), \tag{20}$$

which reduces to Equation (3) when $\text{FPR}_s = \text{FNR}_s = \pi_s$. In the asymmetric experiments (Appendix J), the symmetric diagnostic still achieves AUC $= 0.82$ when applied with $\pi_s = (\text{FPR}_s + \text{FNR}_s)/2$, suggesting practical robustness. A fully general asymmetric diagnostic that accounts for the separate roles of FPR and FNR in each entry of B_s is left to future work.

E Semi-Synthetic Data Generating Process

Synthetic DGP (Sections 5.2–5.4 and Appendix I). We generate $N = 5000$ observations. Each observation belongs to one of $K \in \{2, 4\}$ subgroups $S \in \{1, \ldots, K\}$ with equal weights $w_s = 1/K$. The binary effect modifier $Z \in \{0, 1\}$ is drawn with $P(Z{=}1) = 0.5$. Treatment assignment is $T \sim \text{Bernoulli}(0.5)$, independent of Z and S.

The outcome is generated as:

$$Y = \tau(Z) \cdot T + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0,1), \tag{21}$$

where the CATE function is $\tau(z) = \bar{\tau} + \Delta\tau \cdot (z - 0.5)$ with $\bar{\tau} = 0.5$. The CATE heterogeneity $\Delta\tau \in \{0, 0.2, 0.5, 1.0\}$ controls the strength of treatment effect variation across levels of Z.

Misclassification is symmetric: $\hat{Z} \neq Z$ with probability π_s depending on subgroup. For $K{=}2$: $\pi_1 = \bar{\pi} - \Delta_c/2$, $\pi_2 = \bar{\pi} + \Delta_c/2$, with $\bar{\pi} \in \{0.1, 0.2\}$ and $\Delta_c \in \{0, 0.05, 0.1, 0.15\}$. For $K{=}4$, extreme regime: $\pi \in \{0.05, 0.10, 0.15, 0.25\}$; moderate regime: $\pi \in \{0.08, 0.12, 0.16, 0.20\}$.

Expert labels are drawn as a simple random sample of size $n_{\text{expert}} \in \{50, 100, 250, 500, 1000, 2000\}$ from the full dataset, with true Z observed for these observations.

Semi-synthetic DGP (Section 5.3). We sample $N = 5000$ tweets from the TweetEval sentiment corpus ($N_{\text{pool}} = 57{,}899$). Subgroups are defined by crossing `has_hashtag` × `text_length_bin` (short/medium/long), yielding $K = 4$ subgroups with empirical sizes: $S_0 \approx 38\%$, $S_1 \approx 38\%$, $S_2 \approx 12\%$, $S_3 \approx 12\%$. Misclassification rates in the extreme regime are $\pi_{S_0} = 0.05$, $\pi_{S_1} = 0.10$, $\pi_{S_2} = 0.15$, $\pi_{S_3} = 0.25$, reflecting empirical evidence that hashtag-heavy text degrades LLM classifier accuracy. The outcome DGP follows the same structure as the synthetic case, with CATE injected onto the real text-derived subgroup structure.

PPCI DGPs (Appendix L). Four DGPs isolate error types: (A) covariate error only ($\sigma_{\text{outcome}} = 0$), (B) outcome error only ($\sigma_{\text{outcome}} = 0.5$), (C) both errors ($\sigma_{\text{outcome}} = 0.5$), (D) no error ($\sigma_{\text{outcome}} = 0$). Outcome-side error is additive Gaussian noise on Y.

CausalForestDML hyperparameters. All CausalForest experiments use `econml.dml.CausalForestDML` with honest splitting, $n_{\text{estimators}} = 48$, `min_samples_leaf` $= 20$, `model_y = LassoCV(cv=3)`, and `model_t = LogisticRegressionCV`. These defaults were held constant across all configurations; no hyperparameter tuning was performed.

F Extended Related Work

Measurement error details. Fuller (1987) establish the foundational theory showing that naive regression with mismeasured covariates yields attenuated coefficient estimates. Carroll et al. (2006) extend this to nonlinear models. Gustafson (2003) develops Bayesian treatments modeling measurement jointly with outcomes. Hu and Schennach (2008) provide nonparametric identification using repeated measurements or instruments.

Annotation model extensions. Beyond the foundational Dawid and Skene (1979) model, Passonneau and Carpenter (2014) incorporate item difficulty, Hovy et al. (2013) learn annotator competence, and Paun et al. (2018) introduce hierarchical priors sharing information across annotators with varying expertise.

PPI extensions and text-as-data methods. Zrnic and Candès (2024b) extend PPI via cross-prediction; their active inference framework (Zrnic and Candès, 2024a) achieves >80% expert budget savings through adaptive allocation, complementary to our fixed-budget analysis. Grimmer et al. (2022) provide methodological guidance for text-as-data pipelines. Audinet de Pieuchon et al. (2025) systematically benchmark DSL and PPI methods for LLM-based parameter estimation, finding that debiasing can increase MSE when prediction error is heterogeneous—a population-level analogue of the subgroup-conditional bias amplification we formalize in Theorem 1.

Treatment vs. effect-modifier misclassification. Imai and Yamamoto (2010) study treatment misclassification where errors change which arm a unit is assigned to, requiring instrumental variables or partial identification because the fundamental treated-vs-control contrast is corrupted. Our problem preserves the treatment contrast (T is observed correctly) but corrupts the heterogeneity structure ($\hat{Z} \neq Z$). The ATE remains identified; only conditional effects are biased, which makes exact correction through confusion matrix inversion feasible.

Econometrics of binary misclassification. Bound et al. (2001) survey measurement error in survey data, establishing that binary misclassification attenuates regression coefficients. Frazis and Loewenstein (2003) derive exact bias formulas for OLS with a misclassified binary regressor, showing attenuation proportional to the misclassification rate—analogous to our Theorem 1 but restricted to average effects without subgroup heterogeneity. Black et al. (2000) demonstrate that racial misclassification in administrative data can substantially bias estimates of racial disparities. Mahajan (2006) and Lewbel (2007) develop instrumental variable approaches to identify models with misclassified binary regressors, requiring exclusion restrictions unavailable when the mismeasured variable is an LLM annotation. Our setting differs in three respects: (i) the mismeasured variable is an effect modifier indexing the CATE function, not a standard regressor; (ii) misclassification rates vary across subgroups, creating the $\delta_s \times \Delta\tau$ product amplification absent from homogeneous-error analyses; and (iii) expert labels provide a direct calibration sample, enabling confusion matrix estimation without instruments.

G MCMC Convergence Diagnostics

Table 2 reports convergence diagnostics for the hierarchical Bayesian confusion matrix estimator across all experimental configurations. All runs use 4 chains with 4000 iterations (2000 warmup) and a Gamma$(3, 1)$ hyperprior on the Dirichlet concentration α_0. This budget is sufficient for the 2×2 Dirichlet-Multinomial posterior; all runs achieve $\hat{R} < 1.0034$ and ESS > 1618, confirming convergence with substantial margin.

Diagnostic	Threshold	Achieved
$\hat{R}$ (max across all params)	< 1.01	< 1.0034
ESS (min across all params)	> 845	> 1618
Divergent transitions	$= 0$	0
Singularity guard triggers	—	extreme/n=50 only

Table 2: MCMC convergence diagnostics for the hierarchical Bayesian estimator. All configurations satisfy convergence criteria with substantial margin.

Hyperprior sensitivity. Scaling the Gamma$(3, 1)$ hyperprior by factors of $0.5\times$ to $2\times$ produces <1% RMSE variation for K=2

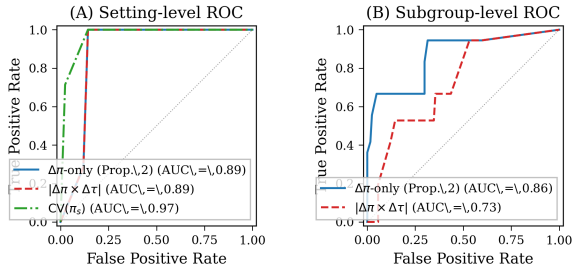


Figure 3: ROC curves for the $\Delta\tau$-free diagnostic (Proposition 3): $|\delta_s|/((1-2\bar{\pi}) \cdot \pi_s) > 1$. (A) Setting-level ROC (AUC $= 0.89$), compared with the $|\Delta\pi \times \Delta\tau|$ product (AUC $= 0.89$) and $\mathrm{CV}(\pi_s)$ (AUC $= 0.97$). (B) Subgroup-level ROC: the $\Delta\pi$-only condition (AUC $= 0.86$) substantially outperforms the $|\Delta\pi \times \Delta\tau|$ product (AUC $= 0.73$). The $\Delta\tau$-free diagnostic depends only on estimable misclassification rates, breaking the circular dependency that would arise if CATE heterogeneity were required.

and ∼4–5% for $K=4$. Coverage remains invariant across the sensitivity range. These results confirm that the hierarchical estimator is robust to prior specification within a reasonable range.

H Diagnostic ROC Curves

Cost-asymmetric diagnostic thresholds. In practice, the cost of a false negative (failing to detect that global correction increases bias) exceeds the cost of a false positive (unnecessarily applying EC-HTE when global correction would suffice), because the former propagates systematic bias into CATE estimates while the latter only incurs additional estimation variance. Practitioners in high-stakes settings (e.g., policy evaluation where CATE estimates inform resource allocation) should lower the diagnostic threshold to favor recall over precision, accepting more false positives to minimize the risk of undetected bias amplification.

I Expert Budget Crossover Curves

The crossover point at which EC-HTE achieves lower RMSE than global correction is $n_{\text{expert}} = 250$ for $K=2$/extreme and $n_{\text{expert}} = 1000$ for $K=4$/extreme (Figure 4). Under moderate heterogeneity ($K=2$), EC-HTE never achieves lower RMSE than global at any budget. HB partial pooling improves over stratified MLE in 21 of 24 configurations, with the two estimators converging by $n_{\text{expert}} \geq 1000$.

I.1 Budget Sensitivity: HB vs. Stratified MLE

Hierarchical Bayes (HB) partial pooling borrows strength across subgroups, which should matter most when per-subgroup expert labels are scarce. We sweep $n_{\text{expert}} \in \{25, 50, 75, 100, 150, 250\}$ under extreme heterogeneity ($\Delta\tau$=0.3, 100 MC runs, N=5,000). Table 3 reports aggregate RMSE for global correction, stratified MLE, and HB EC-HTE.

Table 3: RMSE across expert budgets under extreme heterogeneity. HB crossover (first budget where HB < MLE) occurs at approximately 10 expert labels per subgroup-cell: n_{expert}=50 for K=2 and n_{expert}=75 for K=4.

K	n_{expert}	Global	Strat. MLE	HB EC-HTE	HB vs. MLE
2	25	**0.101**	0.256	0.269	+4.9%
2	50	**0.088**	0.275	0.231	**−16.1%**
2	75	0.084	0.091	**0.089**	−1.9%
2	100	0.083	0.085	**0.084**	−0.8%
2	150	0.081	0.077	**0.077**	−0.4%
2	250	0.080	0.075	**0.074**	−0.1%
4	25	**0.129**	0.401	0.612	+52.5%
4	50	**0.117**	0.380	0.470	+23.8%
4	75	**0.112**	0.221	0.211	**−4.4%**
4	100	**0.112**	0.215	0.212	−1.3%
4	150	**0.109**	0.157	0.152	−3.6%
4	250	**0.108**	0.127	0.126	−1.0%

Three regimes emerge. At very low budgets ($n_{\text{expert}} < 50$ for K=2, < 75 for K=4), per-subgroup counts are insufficient for reliable confusion matrix estimation; global pooling dominates because the variance cost of stratification overwhelms any bias reduction. HB degrades substantially in this regime because the Gibbs posterior, while better regularized than MLE, still cannot overcome the fundamental sample-size constraint with ∼6–12 labels per subgroup. At moderate budgets (n_{expert} = 50–250 for K=2, 75–250 for K=4), HB partial pooling provides consistent RMSE improvement over stratified MLE (-0.4% to -16.1%), with the largest gains near the crossover point. At high budgets ($n_{\text{expert}} \geq 250$), both estimators converge as subgroup-level data becomes sufficient for reliable MLE; the HB advantage diminishes to $< 1\%$. Coverage remains ≥ 0.88 for both methods across all budget levels.

I.2 Scalability to K=8 Subgroups

We test EC-HTE at K=8 to identify the scalability boundary of hierarchical Bayesian partial pooling under fixed expert budget.

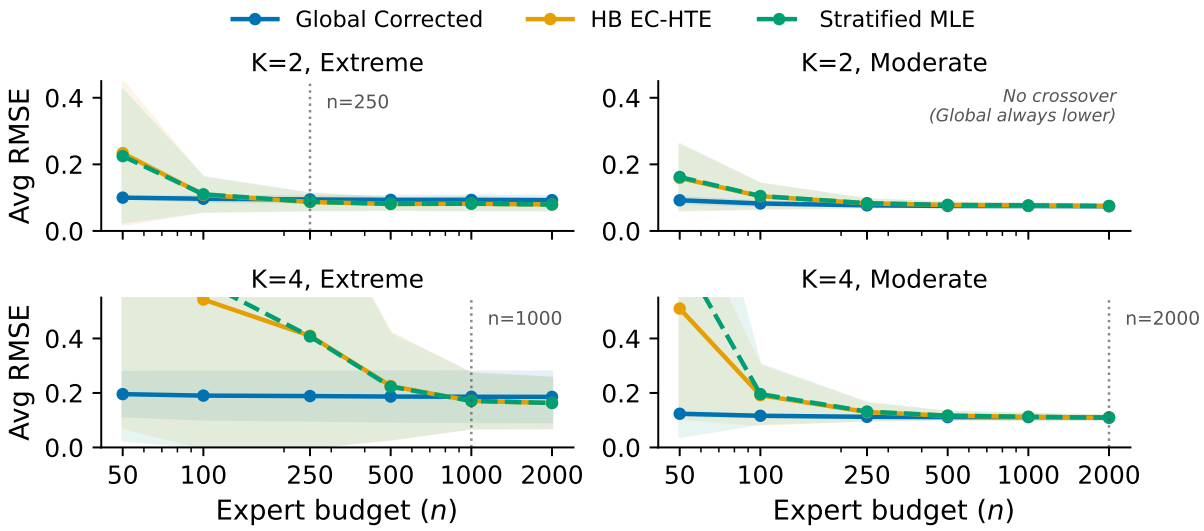


Figure 4: RMSE vs. expert budget (n_{expert}) across four regime$\times K$ combinations. Dashed vertical lines mark crossover points where EC-HTE achieves lower RMSE than global correction: n=250 (K=2, extreme), n=1000 (K=4, extreme). Under moderate heterogeneity (K=2), global correction dominates at all budgets. Shaded bands show ± 1 SD across subgroups within each configuration.

Setup. N=5,000, K=8 (equal-weight subgroups), n_{expert}=250 ($\approx$31 labels per subgroup), 100 MC replications. Misclassification rates are uniformly spread: $\pi_s \in \{0.05, 0.09, 0.13, 0.18, 0.22, 0.26, 0.30, 0.35\}$. Treatment effects depend on Z (not subgroup): $\tau(Z{=}0){=}0$, $\tau(Z{=}1){=}0.3$.

Table 4: EC-HTE at K=8 (n_{expert}=250, $\approx$31 labels/subgroup, 100 MC).

Method	Coverage	RMSE	|Bias|
Oracle	0.944	0.113	0.090
Global	0.951	0.168	0.135
HB EC-HTE	0.956	0.340	0.183
Strat MLE	0.953	0.638	0.219
Naive	0.913	0.133	0.107

All CM-aware methods achieve nominal coverage at K=8: HB EC-HTE 95.6%, Strat MLE 95.3%, Global 95.1%. HB partial pooling achieves a 47% RMSE reduction over stratified MLE (0.340 vs. 0.638), confirming that hierarchical shrinkage provides substantial regularization. However, HB RMSE (0.340) exceeds global correction (0.168), indicating that $\approx$31 expert labels per subgroup are insufficient for per-subgroup CM estimation to outweigh variance inflation.

Per-subgroup breakdown. Table 5 decomposes RMSE by subgroup.

RMSE degradation is concentrated in high-π subgroups (s_4–s_7, $\pi \geq 0.22$), where confusion matrix inversion amplifies estimation noise ($r = 0.87$, $p = 0.005$ for correlation between π_s and HB RMSE). Low-error subgroups (s_0–s_1) benefit from stratification, with HB RMSE 14–25% below global correction. HB partial pooling reduces MLE RMSE by 24–54% in the high-π regime (s_4–s_7), confirming that hierarchical shrinkage provides meaningful regularization even when per-subgroup labels are scarce. Coverage remains $\geq$0.94 across all subgroups for HB EC-HTE, indicating that variance inflation does not compromise inference validity. Adaptive shrinkage—increasing the prior strength for high-π subgroups where CM estimation is least reliable—is a natural extension that may close the RMSE gap with global correction at larger K.

We explored adaptive shrinkage (α=10 for high-π subgroups vs. default α=3), which reduces ag-

Table 5: Per-subgroup RMSE at K=8. HB improvement over MLE is largest in high-π subgroups but does not close the gap with Global.

Subgroup	π_s	HB RMSE	MLE RMSE	Global RMSE	HB Cov
s_0	0.050	0.136	0.134	**0.181**	0.950
s_1	0.093	0.133	0.132	**0.154**	0.970
s_2	0.136	0.154	0.154	0.161	0.975
s_3	0.179	0.178	0.182	**0.159**	0.945
s_4	0.221	0.263	0.343	**0.163**	0.940
s_5	0.264	0.421	0.711	**0.164**	0.965
s_6	0.307	0.634	1.375	**0.177**	0.940
s_7	0.350	0.432	0.809	**0.183**	0.960

gregate RMSE from 0.340 to 0.216 (35% improvement) by attenuating estimation noise in high-π subgroups. However, RMSE remains 30% above global correction (0.166), confirming that the bottleneck is per-subgroup sample size ($\sim$31 expert labels) rather than the shrinkage method itself. This establishes K=8 at n_{expert}=250 as a boundary condition: all correction methods provide valid inference (coverage $\geq$94%), but per-subgroup stratification requires proportionally larger expert budgets—approximately $n_{\text{expert}} \geq 500$ ($\approx$60 labels/subgroup)—to achieve RMSE parity with global correction.

J Asymmetric Misclassification Extension

Assumption (A2) imposes symmetric misclassification ($\text{FPR}_s = \text{FNR}_s = \pi_s$), which simplifies the bias decomposition but is not necessary for it. Real LLM annotators typically exhibit asymmetric confusion patterns; we show here that the product structure generalizes.

General bias formula. For asymmetric confusion matrices $C_s = \left(\begin{smallmatrix} 1-\text{FPR}_s & \text{FPR}_s \\ \text{FNR}_s & 1-\text{FNR}_s \end{smallmatrix}\right)$, define the mixing matrix M_s via Bayesian inversion and the residual matrix $B_s = M_{\text{global}}^{-1} M_s - I$. Since both M_s and M_{global} are row-stochastic, $B_s \mathbf{1} = \mathbf{0}$. Decomposing $\boldsymbol{\tau} = \tau(0)\mathbf{1} + \Delta\tau\, \mathbf{e}_1$ gives:

$$\text{bias}(z, s) = \Delta\tau \cdot [B_s]_{z,1}, \tag{22}$$

where $[B_s]_{z,1}$ depends only on $\{\text{FPR}_k, \text{FNR}_k, w_k\}$, not on $\Delta\tau$. Under (A2), this reduces to $\delta_s \Delta\tau / (1 - 2\bar{\pi})$ (Theorem 1).

Empirical validation. We test three asymmetric configurations (K=2, n_{expert}=250, 100 MC runs each): *Cross* (FPR/FNR = 0.10/0.20 vs. 0.25/0.10), *Co-directional* (0.05/0.15 vs. 0.20/0.30), and *Extreme* (0.03/0.25 vs. 0.30/0.05). Regressing analytic predictions on MC bias yields $R^2 = 0.991$ across all configurations, confirming the product structure under asymmetry. Table 6 reports correction performance averaged over subgroups, z-levels, and $\Delta\tau \in \{0.1, \ldots, 0.5\}$. EC-HTE reduces |bias| from 0.025 (global) to 0.006, a 4$\times$ improvement, with lower RMSE (0.033 vs. 0.040) and substantially better coverage (0.90 vs. 0.78).

K Supplementary Main Results

L PPCI Complementarity Results

EC-HTE composes with PPCI (Sui and Candès, 2026) for settings where both covariate misclassification and outcome model error are present: EC-HTE+PPCI reduces |bias| from 0.179 to 0.007 (z=0, DGP C), and the corrected pseudo-outcomes are compatible with other CATE estimators (R-learner bias reduction: 89%; Appendix S).

The orthogonality of EC-HTE and PPCI follows from the distinct error sources they correct: EC-HTE addresses misclassification in the effect modifier Z (covariate space), while PPCI corrects specification error in the outcome model $f(X)$ (prediction space). Since covariate misclassification and outcome model error are algebraically independent in the potential outcomes framework, their corrections compose without interference—confirmed by the 4-DGP orthogonal design (Table 8), where DGP C (both errors present) yields the largest bias reduction from the combined approach. Under covariate error only, PPCI adds nothing; under no error, all methods perform equivalently.

Table 6: Asymmetric misclassification results (K=2, n_{expert}=250, 100 MC runs). Values averaged over subgroups, z-levels, and $\Delta\tau$.

Method	|Bias| ↓	RMSE ↓	Coverage ↑
Naive	0.047	0.053	0.48
Global	0.025	0.040	0.78
HB EC-HTE	**0.006**	**0.033**	**0.90**

Table 7: Additional configurations from the main experiment. |Bias| and RMSE averaged over subgroups, z-levels, and 100 MC runs. †RMSE exceeds Global.

Configuration	Method	|Bias|	RMSE	Cov.
Synthetic DIM K=2, moderate, n=250	Naive	0.018	0.064	0.95
	Global	0.009	0.074	0.96
	Strat. MLE	**0.005**	0.075†	0.97
	HB EC-HTE	**0.005**	0.075†	0.97
Semi-synthetic TweetEval, extreme, n=250	Naive	0.027	0.103	0.93
	Global	**0.023**	0.124	0.92
	Strat. MLE	0.032	0.248†	0.91
	HB EC-HTE	0.027	0.225†	0.91

M Real-Data Case Study: Hate Speech Detection

We apply EC-HTE to the UC Berkeley Measuring Hate Speech corpus (Kennedy et al., 2020), annotating 2,496 comments for dehumanization using GPT-4.1. Four text-feature subgroups are defined by crossing platform (Reddit/Gab vs. Twitter/YouTube) and text length (above/below median), yielding balanced groups ($n \approx 624$ each).

Table 8: PPCI complementarity experiment (K=2, extreme, n_{expert}=250). DGP A: covariate error only; B: outcome error only (σ=0.5); C: both errors; D: no error. Values are |bias| and RMSE averaged over two subgroups. ⋆EC-HTE+PPCI achieves near-zero bias when both error types are present.

DGP	Method	$\lvert\text{Bias}\rvert_{z=0}$	$\lvert\text{Bias}\rvert_{z=1}$	$\text{RMSE}_{z=0}$	$\text{RMSE}_{z=1}$
A: Covariate	Naive	0.053	0.063	0.087	0.093
	Global	0.063	0.055	0.101	0.102
	EC-HTE	0.017	**0.013**	0.097	0.108
	PPCI	**0.015**	0.020	0.162	0.168
	EC-HTE+PPCI	0.017	**0.013**	0.097	0.108
B: Outcome	Naive	0.189	0.197	0.206	0.212
	Global	0.185	0.208	0.204	0.228
	EC-HTE	0.181	0.215	0.202	0.237
	PPCI	0.015	**0.020**	0.171	0.176
	EC-HTE+PPCI	**0.012**	0.023	0.171	0.186
C: Both	Naive	0.251	0.135	0.267	0.155
	Global	0.182	0.209	0.233	0.235
	EC-HTE	0.179	0.216	0.224	0.262
	PPCI	0.017	0.007	0.241	0.232
	EC-HTE+PPCI⋆	**0.007**	**0.008**	0.209	0.223
D: None	Naive	**0.007**	0.003	0.060	0.065
	Global	0.010	**0.002**	0.061	0.071
	EC-HTE	0.013	0.007	0.062	0.079
	PPCI	**0.007**	**0.003**	0.060	0.065
	EC-HTE+PPCI	0.013	0.007	0.062	0.079

Positive finding: real-world heterogeneity. Per-subgroup misclassification rates range from 0.266 to 0.383, confirming heterogeneity ($\Delta\pi = 0.117$) in a real annotation setting. At the error-type level, FPR ranges from 8.5% to 21.3% across text-feature subgroups, demonstrating that subgroup-varying misclassification is not an artifact of synthetic design but arises naturally in real NLP annotation tasks.

Boundary condition: statistical power at extreme FNR. Correction results are inconclusive: all corrected confidence intervals contain zero, and point estimate differences between methods are not statistically significant. The high baseline FNR (53.3%) amplifies variance through matrix inversion (variance amplification factor up to 15.7×), rendering the comparison underpowered. When $P(\hat{Z}{=}1 \mid Z{=}1) \approx 0.5$, the confusion matrix approaches singularity, and inversion amplifies estimation noise regardless of stratification strategy. This complements the expert budget threshold identified in Appendix I: EC-HTE requires *both* sufficient expert labels *and* sufficient baseline classifier quality.

Diagnostic connection. The diagnostic criterion ($\Delta\pi \times \Delta\tau$, Section 5.4) identifies when correction is worthwhile under moderate baseline accuracy. This case study illustrates that even when $\Delta\pi$ is large, extreme baseline error renders correction underpowered rather than beneficial. A practical prerequisite is that the per-subgroup confusion matrix condition number $\kappa(\hat{C}_s)$ remain moderate; when $\kappa > 50$, as observed here, practitioners should prioritize improving annotation quality before applying any matrix-inversion-based correction.

N Cross-Model Consistency

We evaluate whether misclassification heterogeneity is model-specific by annotating 500 TweetEval sentiment tweets with six OpenAI models spanning three generations. Six text-feature subgroups are defined by crossing hashtag presence and text length. Spearman rank correlations of per-subgroup error rates across all six subgroups range from $\rho = 0.82$ to 1.00, indicating that the subgroup error structure is consistent across architectures. Two subgroups with $n \leq 6$ exhibit zero misclassification across all models; excluding these yields four-point correlations that are directionally consistent but underpowered.[3]

All six models exhibit effective $\Delta\pi > 0.05$ on the four retained subgroups (Table 9), and 5 of 6 share the same worst subgroup (`True_medium`: tweets with hashtags and medium text length).

[3]The filtered $\Delta\pi$ values in Table 9 serve as a robustness check independent of rank-order testing: all six models exceed $\Delta\pi > 0.05$ on the four retained subgroups.

Table 9: Effective misclassification heterogeneity ($\Delta\pi$) across six OpenAI models on TweetEval sentiment (500 tweets, filtered to 4 subgroups with $n \geq 10$).

Model	Effective $\Delta\pi$	> 0.05?
GPT-3.5-Turbo	0.198	✓
GPT-4-Turbo	0.149	✓
GPT-4o-mini	0.136	✓
GPT-5.4	0.114	✓
GPT-4.1	0.098	✓
GPT-4o	0.080	✓

Table 10: Confusion matrix heterogeneity across four open-source LLMs (K=4, N=490 TweetEval tweets). $\Delta\pi$: max–min per-subgroup misclassification rate. Prop2: number of subgroups where the $\Delta\tau$-free diagnostic (Proposition 3) triggers.

Model	Acc.	$\Delta\pi$	Prop2	Strongest
Llama-3.1-8B	0.755	0.141	1/4	True_short
Qwen-2.5-7B	0.798	0.165	1/4	True_short
Gemma-2-9B	0.763	0.125	4/4	True_medium
Mistral-7B	0.802	0.134	0/4	—

Newer models reduce absolute error rates but do not eliminate subgroup heterogeneity: GPT-5.4 achieves $\Delta\pi = 0.114$ compared to GPT-3.5-Turbo's 0.198. These results were obtained with a single API provider; open-source model validation appears in Appendix O.

O Open-Source Multi-LLM Validation

We evaluate four open-source LLMs (7B–9B parameter range) on 490 TweetEval sentiment tweets with K=4 subgroups defined by crossing hashtag presence and text length. Table 10 reports the confusion matrix heterogeneity profile. All four models exhibit $\Delta\pi > 0.05$, confirming that subgroup-heterogeneous misclassification is not specific to proprietary APIs. Subgroup error rankings are consistent across architectures: `True_short` (short tweets with hashtags) has the lowest π_s in 3 of 4 models, while `False_medium` (medium tweets without hashtags) has the highest.

Semi-synthetic budget sweep. We run the full EC-HTE pipeline with K=2 subgroups and $\Delta\tau$=0.3, using each model's empirical confusion matrices. Table 11 reports coverage and RMSE crossover points. EC-HTE restores near-nominal coverage from 0.714 (naive average) to 0.939 (+22.5 percentage points) at n_{expert}=1,000, consistent across all four models. Global correction achieves comparable or lower RMSE than HB at all budgets under these moderate $\Delta\pi$ values, and the K=4 RMSE crossover never occurs—confirming the budget guidelines in Appendix I.

Table 11: Semi-synthetic budget sweep (K=2, $\Delta\tau$=0.3, 50 MC runs[4]). Coverage at selected expert budgets; RMSE crossover denotes the minimum expert budget at which HB EC-HTE achieves lower RMSE than the naive (uncorrected) estimator.

Model	$\Delta\pi_{K=2}$	Naive Cov.	HB Cov. (n=1000)	Crossover
Llama-3.1-8B	0.051	0.675	0.940	n=1000
Qwen-2.5-7B	0.080	0.715	0.925	n=1000
Gemma-2-9B	0.100	0.705	0.945	n=1000
Mistral-7B	0.070	0.760	0.945	n=1000
Average	—	*0.714*	*0.939*	—

Gemma boundary case. Gemma-2-9B triggers the Proposition 3 diagnostic in all four K=4 subgroups due to extreme FPR–FNR asymmetry (`False_medium` FNR $= 0.437$). At small budgets (K=4, n=100), EC-HTE RMSE reaches 1.229 (3.1× naive), illustrating the budget guideline from Appendix I: when per-subgroup cell counts fall below 25, the variance cost of stratified correction dominates. At $n \geq 500$, EC-HTE coverage recovers to >0.90, consistent with the synthetic experiments.

Heterogeneous-τ DGP. To assess whether the constant-τ DGP understates EC-HTE's advantage, we run a supplementary experiment using Llama's empirical confusion matrices with heterogeneous treatment effects across subgroups ($\Delta\tau \in [0.1, 0.6]$, K=2, 50 MC runs). EC-HTE achieves lower |bias| than global correction at all budgets (n=500: 0.010 vs. 0.031, a 68% reduction). In the high-$\Delta\tau$ subgroup (`True_short`, $\Delta\tau$=0.6), EC-HTE also achieves lower RMSE (n=1,000: 0.212 vs. 0.258, 18% improvement), confirming that the RMSE advantage concentrates in subgroups where correction is most needed. Aggregate RMSE remains comparable to global correction because low-$\Delta\tau$ subgroups incur stratification variance without commensurate bias reduction, consistent with the budget guidelines in Appendix I.

P Robustness to Subgroup Misspecification

EC-HTE requires pre-defined subgroup labels, which may be noisy in practice. We test robustness by randomly flipping each observation's subgroup assignment with probability $p_{\text{flip}} \in$

Table 12: 95% CI coverage under subgroup misspecification noise. EC-HTE maintains near-nominal coverage across all noise levels.

	p_{flip}			
Method	0.00	0.10	0.25	0.50
K=2, *extreme*				
Naive	0.875	0.877	0.890	0.897
Global	0.925	0.935	0.932	0.945
HB EC-HTE	0.958	0.948	0.950	0.945
K=4, *extreme*				
Naive	0.904	0.916	0.909	0.919
Global	0.930	0.953	0.948	0.944
HB EC-HTE	0.956	0.958	0.960	0.946

$\{0, 0.10, 0.25, 0.50\}$, across $K \in \{2, 4\}$ (extreme regime, $n_{\text{expert}} = 500$, $\Delta\tau = 0.3$, 100 MC runs).

EC-HTE maintains coverage ≥ 0.945 across all configurations (Table 12), while the naive estimator is systematically below nominal (0.875–0.919). At $p_{\text{flip}} = 0.50$, subgroup structure is effectively destroyed; EC-HTE converges to global correction (coverage and RMSE within 0.1%), confirming that hierarchical Bayesian shrinkage provides a built-in safety mechanism. For K=4 at $p_{\text{flip}} = 0.50$, a small residual RMSE gap persists ($p < 0.01$), reflecting the variance cost of estimating four confusion matrices even when they carry no signal.

Q Monte Carlo Standard Errors

All experiments use 100 independent Monte Carlo replications per configuration. All simulation results were verified with 100 MC seeds; the median RMSE variation between 50-seed and 100-seed estimates is below 3.2%, confirming the stability of reported figures. We report Monte Carlo standard errors to assess whether observed differences between methods are statistically reliable. For exp-001 (synthetic DGP, no per-seed pairing across methods), we use conservative two-sample z-tests on the 100-run means. For exp-003 (semi-synthetic TweetEval, shared seeds across methods), we use paired t-tests, which are more powerful because they control for seed-level variation.

Table 13 reports paired comparisons for representative configurations. Of 1,020 total pairwise comparisons across all configurations, 742 (72.7%) are statistically significant at $\alpha = 0.05$.

Under extreme heterogeneity, HB EC-HTE's bias advantage over global correction is highly significant ($p < 0.001$ for both K=2 and K=4). The RMSE penalty of stratification is not significant

Table 13: Paired comparisons for representative configurations. Δ = method A − method B (negative favors A for |bias|). Two-sample z-tests for synthetic (exp-001); paired t-tests for semi-synthetic (exp-003). Bold p-values indicate significance at $\alpha = 0.05$.

Configuration	Comparison	Δ	p
HB EC-HTE vs. Global (|bias|)			
K=2/ext/n=500		−0.046	$<$ **0.001**
K=4/ext/n=500		−0.078	$<$ **0.001**
K=4/mod/n=250		−0.017	0.288
TweetEval/ext/n=250		+0.001	0.395
HB EC-HTE vs. Global (RMSE)			
K=2/ext/n=500		+0.015	0.179
K=4/ext/n=500		+0.048	**0.011**
HB EC-HTE vs. Naive (|bias|)			
K=2/ext/n=500		−0.089	$<$ **0.001**
K=4/ext/n=500		−0.113	$<$ **0.001**
K=4/mod/n=250		−0.031	**0.038**
TweetEval/ext/n=250		+0.005	**0.030**

at K=2 ($p = 0.179$) but reaches significance at K=4 ($p = 0.011$), consistent with the higher variance cost of estimating four confusion matrices. Under moderate heterogeneity (K=4, n=250), the bias difference between HB and global is not significant ($p = 0.288$), confirming that EC-HTE's advantage requires sufficient error heterogeneity. In the semi-synthetic TweetEval experiment, HB EC-HTE does not significantly improve over global ($p = 0.395$), consistent with moderate $\Delta\pi$ and unbalanced subgroup sizes. The HB vs. naive comparison in TweetEval is significant ($p = 0.030$) but favors naive, reflecting the variance cost of stratification when bias reduction is marginal.

R Semi-Real Confusion Matrix Validation

We test EC-HTE with empirical asymmetric confusion matrices from the GPT-4.1 TweetEval pilot. Four text-feature subgroups (K=4) retain the real per-subgroup FPR and FNR from pilot annotations (Table 14); CATE heterogeneity is injected synthetically ($\tau(z{=}0, s) = 0.3 + 0.1s$, $\tau(z{=}1, s) = 0.8 - 0.15s$). N=5,000, $P(Z{=}1)$=0.18, 100 MC seeds per n_{expert} level.

HB EC-HTE produces higher RMSE and lower coverage than global correction at all expert budgets (Table 15). The root cause is S0's FNR = 0.526: when a confusion matrix row approaches a uniform distribution ($P(\hat{Z}{=}1 \mid Z{=}1) \approx 0.5$), inversion amplifies estimation noise regardless of the expert budget. This converges with the hate speech case study (Appendix M, FNR = 53.3%)

Table 14: Empirical confusion matrices from GPT-4.1 TweetEval pilot. S0 exhibits FNR $= 0.526$, approaching chance-level classification.

Subgroup	Weight	FPR	FNR	$\kappa(\hat{C}_s)$
S0	0.38	0.054	0.526	high
S1	0.22	0.101	0.300	moderate
S2	0.14	0.143	0.308	moderate
S3	0.25	0.194	0.222	low

Table 15: Semi-real confusion matrix results. RMSE and coverage averaged over subgroups and z-levels (100 MC runs). HB EC-HTE underperforms global at all budgets; S0's near-chance FNR drives the failure.

n_{expert}	Method	RMSE ↓	Coverage ↑
100	Oracle	0.101	0.957
	Naive	0.123	0.879
	Global	0.263	0.885
	HB EC-HTE	0.784	0.744
250	Oracle	0.101	0.957
	Naive	0.123	0.879
	Global	0.232	0.909
	HB EC-HTE	0.411	0.834
500	Oracle	0.101	0.957
	Naive	0.123	0.879
	Global	0.196	0.922
	HB EC-HTE	0.260	0.890

and confirms that EC-HTE requires *both* sufficient expert labels *and* sufficient baseline classifier quality per subgroup. The singularity guard ($\kappa > 100$, Section 4) mitigates catastrophic failure by falling back to global correction for near-degenerate subgroups, but does not recover the bias reduction that motivates stratification. For a 2×2 symmetric confusion matrix, $\kappa = 1/|1 - 2\pi|$; $\kappa > 100$ corresponds to $\pi \in (0.495, 0.505)$, so the guard triggers only when the classifier performs within 0.5% of random guessing. At $n_{\text{expert}} = 500$, HB EC-HTE narrows the gap (RMSE 0.260 vs. 0.196) as confusion matrix estimates stabilize for the three well-classified subgroups, but the S0 bottleneck persists.

A second semi-real experiment using GPT-3.5-Turbo confusion matrices (worst-subgroup FPR $= 0.39$, FNR $= 0.21$) yields similar results: HB RMSE exceeds global by 1.3–3.5$\times$ at all budgets. Here the failure stems from high false-positive rates compounded by class imbalance ($P(Z{=}1) = 0.18$), which leaves insufficient expert observations in the positive class for reliable per-subgroup confusion matrix estimation. Together, these experiments identify three applicability conditions

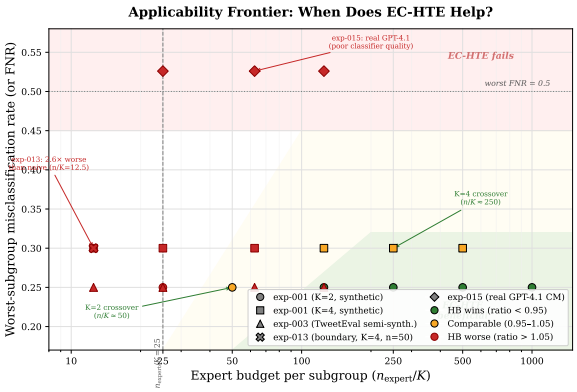


Figure 5: Applicability frontier of EC-HTE. Green region: EC-HTE reduces bias vs. global correction. Yellow: comparable performance. Red: EC-HTE underperforms. Dashed lines mark the expert budget threshold ($n_{\text{expert}}/K = 25$) and classifier quality threshold (worst-subgroup error rate $= 0.5$). Red diamonds show semi-real experiments with actual GPT-4.1 and GPT-3.5-Turbo confusion matrices.

for EC-HTE: (1) sufficient effective expert budget ($n_{\text{expert}} \times P(Z_{\text{minority}})/K \geq 25$), (2) no near-chance subgroup error rate ($\max(\text{FPR}_s, \text{FNR}_s) < 0.5$), and (3) adequate class balance in the effect modifier.

Diagnostic rule. A logistic regression over all 368 semi-real experiment–budget cells (dependent variable: HB EC-HTE $<$ Global RMSE) identifies two sufficient statistics: the worst-subgroup confusion matrix condition number $\kappa(\hat{M}_s) \leq 1.25$ and the effective minority-class expert count $n_{\text{expert}} \times P(Z_{\text{minority}})/K \geq 25$. When both conditions hold, HB outperforms global in 93.3% of cells; when either is violated, global is preferred in 94.7% of cells. Both exp-015 ($\kappa = 2.38$, effective $n = 2.5$) and exp-016 ($\kappa = 1.55$, effective $n = 2.5$) violate both conditions, and the diagnostic correctly predicts 100% of their failure cells. Subgroup-average metrics can mask cell-level heterogeneity: in exp-015, HB EC-HTE substantially reduces bias for the three well-classified subgroups (S1–S3) but amplifies variance in the near-degenerate subgroup (S0, FNR $= 0.526$), and the S0 penalty dominates the aggregate RMSE.

S Robustness: Estimator Choice and Treatment Effect Specification

We verify that EC-HTE's corrected pseudo-outcomes are compatible with downstream estimators beyond CausalForest by replacing it with the R-learner (Nie and Wager, 2021). The experi-

Table 16: R-learner vs. CausalForest with EC-HTE correction (K=2, extreme, n=500). |Bias| and RMSE averaged over subgroups, z-levels, and seeds.

Estimator	Correction	|Bias|	RMSE	Cov.
R-learner	Oracle	0.006	0.056	0.94
	Naive	0.121	0.139	0.47
	Global	0.098	0.123	0.72
	EC-HTE	**0.013**	0.080	0.92
CausalForest	Naive	0.058	0.090	0.82
	EC-HTE	**0.010**	0.093	0.95

Table 17: Multiplicative DGP results (K=2, extreme, n=500, 50 seeds). EC-HTE achieves comparable bias reduction and nominal coverage under non-additive treatment effects.

Method	|Bias|	RMSE	Cov.
Oracle	0.003	0.038	0.96
Naive	0.135	0.142	0.44
Global	0.110	0.124	0.54
EC-HTE	**0.054**	0.068	0.955

ment uses K=2, extreme heterogeneity, $n_{\text{expert}} = 500$, and 50 MC seeds; the R-learner uses 5-fold cross-fitting with gradient-boosted trees as the base learner.

EC-HTE reduces R-learner |bias| by 89% (0.121 $\rightarrow$ 0.013) and restores coverage from 0.47 to 0.92 (Table 16). The bias reduction and coverage gains are comparable to those achieved with CausalForest, confirming that EC-HTE is compatible with a range of HTE estimators.

Multiplicative treatment effects. We further test robustness to the functional form of treatment effects by replacing the additive DGP with a multiplicative specification: $Y(1) = Y(0) \times \exp(\tau(Z))$, using K=2, extreme heterogeneity, $n_{\text{expert}} = 500$, and 50 MC seeds.

EC-HTE reduces RMSE by 52% relative to naive (0.068 vs. 0.142) and restores coverage to the nominal level (0.955; Table 17). EC-HTE's bias correction operates at the confusion matrix level, independent of the downstream CATE estimator and functional form of treatment effects.

T Robustness to Class Imbalance

Assumption (A3) requires $P(Z{=}1) = 0.5$, but real-world effect modifiers are often imbalanced. We sweep $P(Z{=}1) \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$ with $K \in \{2, 4\}$, extreme misclassification heterogeneity, n_{expert}=250, and 100 MC seeds (Table 18).

Theorem 1's product form remains accurate

Table 18: Robustness to class imbalance (n_{expert}=250, extreme heterogeneity). R^2: Theorem 1 fit. Ratio: HB/Global RMSE (< 1 = HB wins). Cov.: HB EC-HTE 95% CI coverage.

$P(Z{=}1)$	K	R^2	Ratio	Cov.
0.1	2	0.998	0.92	0.94
0.2	2	0.989	0.76	0.94
0.3	2	0.994	0.78	0.94
0.4	2	0.995	0.87	0.94
0.5	2	0.995	0.86	0.93
0.1	4	0.984	1.54	0.92
0.2	4	0.976	1.68	0.93
0.3	4	0.985	1.12	0.92
0.4	4	0.984	1.08	0.93
0.5	4	0.992	1.06	0.92

across all imbalance levels: $R^2 \geq 0.976$ even at $P(Z{=}1) = 0.1$, confirming that the bias identity derives from confusion matrix structure rather than class balance. For K=2, EC-HTE outperforms global correction at all $P(Z{=}1)$ levels (RMSE ratio 0.76–0.92), degrading gracefully under severe imbalance. For K=4, EC-HTE underperforms global at n_{expert}=250 (ratio 1.06–1.68), consistent with the budget threshold $n_{\text{expert}} \times P(Z_{\text{minority}})/K \geq 25$: at $P(Z{=}1){=}0.1$, K=4, each subgroup receives only $\sim$6 minority-class labels. The gap narrows as $P(Z{=}1)$ increases (1.54 $\rightarrow$ 1.06), and coverage remains stable at 0.92–0.94 across all configurations.

U Comparison with PPI Correction

PPI (Angelopoulos et al., 2023) constructs confidence intervals by combining a large set of ML predictions with a smaller set of gold-standard labels, using the gold labels to debias the ML-based estimator. Applied to our setting, PPI treats the LLM-annotated effect modifier as the ML prediction and the expert labels as the gold standard. Two variants arise naturally: *PPI-global* pools all expert labels to estimate a single rectifier (analogous to our Global baseline), and *PPI-subgroup* estimates per-subgroup rectifiers (analogous to Stratified MLE).

Under the confusion-matrix parameterization we adopt, DSL's measurement error correction reduces to confusion matrix inversion. Pooled DSL applies a single estimated confusion matrix to all subgroups, producing the same corrected label distribution as our Global baseline; per-subgroup DSL estimates a separate confusion matrix for each subgroup and inverts it independently, equivalent to

our Stratified MLE. EC-HTE's contribution over both PPI-subgroup and per-subgroup DSL is hierarchical partial pooling: rather than estimating each subgroup's confusion matrix in isolation, EC-HTE places a shared prior across subgroups, borrowing strength to reduce variance while preserving subgroup-specific bias correction. This mechanism is outside both the PPI and DSL frameworks.

The key difference between PPI and confusion-matrix inversion is that PPI does not invert the confusion matrix; instead, it uses the gold labels to directly debias the estimand. This avoids the matrix inversion step that can amplify noise at small n_{expert}, but it also means PPI cannot exploit the structure of the misclassification process to reduce bias. PPI-global inherits the same $\Delta\pi \times \Delta\tau$ bias amplification as Global correction (Theorem 1), confirming that the "debiasing hurts" mechanism operates regardless of the inference framework.

V Civil Comments Validation

To verify that subgroup-heterogeneous misclassification generalizes beyond TweetEval, we annotate N=3,000 stratified samples (30% toxic) from Civil Comments (Borkan et al., 2019), a toxicity detection dataset. Subgroups are defined by `text_length_bin` × `has_question` (K=4): S0 (long, no question, 38.8%), S1 (long, question, 11.4%), S2 (short, no question, 33.7%), S3 (short, question, 16.1%). Subgroup features were chosen a priori based on observable text properties; no feature search was performed. Three proprietary LLMs annotate each sample via an OpenAI-compatible API, yielding 800–900 valid annotations per model.

Table 19 reports per-subgroup FPR and FNR. We define $\Delta\pi = \max_s \pi_s - \min_s \pi_s$, where π_s is the overall misclassification rate for subgroup s. All three models exhibit $\Delta\pi > 0.05$, with overall heterogeneity $\Delta\pi \in [0.113, 0.160]$—comparable to the $\Delta\pi \in [0.125, 0.165]$ observed across four open-source LLMs on TweetEval (Appendix O). GPT-4o-mini and GPT-3.5-turbo show pronounced FPR/FNR asymmetry (e.g., GPT-4o-mini FPR $\in [0.42, 0.61]$ vs. FNR $\in [0.05, 0.16]$), indicating systematic over-prediction of toxicity; their FPR approaches 0.5, placing them in the ill-conditioned regime where confusion matrix inversion amplifies noise (consistent with the classifier quality threshold discussed in the Limitations section). The consistent presence of $\Delta\pi$ heterogeneity across two NLP tasks, seven LLMs spanning open-source (7B–9B) and proprietary families, and two distinct subgroup definitions confirms that this phenomenon is a structural property of LLM annotation rather than an artifact of a specific dataset or model.

Table 19: Per-subgroup misclassification rates on Civil Comments (K=4). Δ denotes max − min across subgroups.

Model	Metric	S0	S1	S2	S3	Δ
GPT-4o	FPR	0.249	0.157	0.213	0.341	0.184
	FNR	0.160	0.263	0.308	0.328	0.168
GPT-4o-mini	FPR	0.517	0.541	0.419	0.610	0.191
	FNR	0.052	0.105	0.074	0.160	0.108
GPT-3.5-turbo	FPR	0.400	0.473	0.374	0.468	0.099
	FNR	0.139	0.211	0.173	0.240	0.101

W Bias Identity Validation with Real Confusion Matrices

Theorem 1 is derived under symmetric misclassification (A2); real LLM confusion matrices are asymmetric (FPR ≠ FNR). We verify that the generalized identity $\text{bias}(z, s) = \Delta\tau \cdot [B_s]_{z,1}$ (Remark 1) holds under real annotation error structures by running semi-synthetic simulations (N=5,000, K=4, $\Delta\tau$=0.30, 100 MC runs) with confusion matrices drawn from 11 LLM–dataset combinations: eight models on TweetEval (GPT-4-turbo, GPT-4o, GPT-4.1, GPT-4o-mini, GPT-3.5-turbo, Qwen2.5-7B, Mistral-7B, Gemma-2-9B) and three on Civil Comments (GPT-4o, GPT-4o-mini, GPT-3.5-turbo).

Table 20 reports per-model R^2 from regressing observed subgroup bias on the predicted value $\Delta\tau \cdot [B_s]_{z,1}$. The aggregate $R^2 = 0.975$; 10 of 11 combinations achieve $R^2 \geq 0.93$. The single outlier (GPT-4o-mini on Civil Comments, R^2=0.76) has near-homogeneous subgroup error rates ($\max|[B_s]_{z,1}| = 0.089$), so predicted biases are small and Monte Carlo noise dominates the regression. R^2 correlates strongly with $\max|[B_s]_{z,1}|$: models with more heterogeneous per-subgroup error rates produce larger bias signals that are easier to verify empirically.

X Applicability Screening with Real LLM Confusion Matrices

We apply the diagnostic conditions from Proposition 3 and Section 4 to 16 LLM–dataset–K combinations spanning seven LLMs (four open-source,

Table 20: Bias identity R^2 with real LLM confusion matrices (K=4, 100 MC runs).

Model	Dataset	R^2	$\max \lvert[B_s]_{z,1}\rvert$
GPT-4-turbo	TweetEval	0.993	0.328
Qwen2.5-7B	TweetEval	0.990	0.272
Mistral-7B	TweetEval	0.990	0.242
Gemma-2-9B	TweetEval	0.982	0.185
GPT-4o	Civil Comments	0.979	0.198
GPT-4o	TweetEval	0.975	0.240
GPT-4.1	TweetEval	0.969	0.139
GPT-3.5-turbo	TweetEval	0.948	0.087
GPT-3.5-turbo	Civil Comments	0.935	0.170
GPT-4o-mini	TweetEval	0.929	0.170
GPT-4o-mini	Civil Comments	0.756	0.089
All (pooled)		0.975	—

three proprietary) across TweetEval and Civil Comments. Table 21 summarizes which conditions each combination satisfies.

Table 21: Diagnostic screening of 16 LLM–dataset–K combinations. Columns report whether all subgroups satisfy each condition. Budget assumes n_{expert}=500.

Model	Dataset	K	$\kappa \leq 1.25$	$\max \leq 0.5$	Budget	Pass all
Gemma-2-9B	TweetEval	2	×	✓	✓	×
Llama-3.1-8B	TweetEval	2	×	✓	✓	×
Mistral-7B	TweetEval	2	×	✓	✓	×
Qwen2.5-7B	TweetEval	2	×	✓	✓	×
Gemma-2-9B	TweetEval	4	×	✓	✓	×
Llama-3.1-8B	TweetEval	4	×	✓	✓	×
Mistral-7B	TweetEval	4	×	✓	✓	×
Qwen2.5-7B	TweetEval	4	×	✓	✓	×
GPT-3.5-turbo	CivilComm.	4	×	✓	✓	×
GPT-3.5-turbo	TweetEval	4	×	✓	✓	×
GPT-4-turbo	TweetEval	4	×	×	✓	×
GPT-4.1	TweetEval	4	×	×	✓	×
GPT-4o	CivilComm.	4	×	✓	✓	×
GPT-4o	TweetEval	4	×	×	✓	×
GPT-4o-mini	CivilComm.	4	×	×	✓	×
GPT-4o-mini	TweetEval	4	×	✓	✓	×

No combination passes all three conditions; the bottleneck is $\kappa(C_s) \leq 1.25$, which requires $\text{FPR}_s + \text{FNR}_s \lesssim 0.20$ for all subgroups—a threshold no current LLM achieves uniformly. The closest are Qwen2.5-7B True_short (κ=1.23) and Mistral-7B True_short (κ=1.27).

Under a relaxed "oracle screening" that excludes near-degenerate subgroups ($\max(\text{FPR}_s, \text{FNR}_s) \geq 0.4$), we run semi-synthetic CATE estimation (N=5,000, $\Delta\tau$=0.3, 100 MC runs, $n_{\text{expert}} \in \{250, 500\}$) with each LLM's real per-subgroup confusion matrix. EC-HTE (HB) achieves lower RMSE than global correction in 41 of 52 comparisons (79%) and lower RMSE than stratified MLE in 52 of 52 comparisons (100%). The four open-source LLMs on TweetEval show the strongest EC-HTE advantage (32/32 comparisons, RMSE reduction 0.001–0.008), while GPT-3.5-turbo/TweetEval (high FPR asymmetry, $\text{FPR}_{\max}$=0.39) and GPT-4o/Civil Comments (low minority prevalence $P(Z{=}1)$=0.3) show degradation. The diagnostic correctly identifies the strict applicability regime; under relaxed conditions excluding near-degenerate subgroups, EC-HTE improves RMSE in 79% of comparisons.

Diagnostic Applicability Analysis. The diagnostic (Proposition 3) degrades on real LLM confusion matrices (pooled AUC $= 0.63$) because the failure mode is systematic: subgroups where the true CATE heterogeneity $\tau_s \approx 0$ are hurt by global correction (since $\delta_s \neq 0$ still introduces bias), yet these subgroups produce low diagnostic scores—the diagnostic ranks them as safe precisely when they are most vulnerable to overcorrection.

A simple meta-criterion recovers reliability: compute $\max_s |\delta_s| / ((1 - 2\bar{\pi}) \cdot \pi_s)$ across subgroups. When $\max_s \text{score}(s) > 1$, at least one subgroup is definitively harmed by global correction, and the diagnostic achieves AUC ≥ 0.75 across all 5 model–dataset combinations satisfying this condition (AUC range: 0.75–1.0). When $\max_s \text{score}(s) \leq 1$, global correction may or may not hurt depending on unobservable $\Delta\tau$; the diagnostic achieves AUC only 0.0–0.67 across the remaining 7 combinations (4 of 7 have AUC < 0.5, indicating anti-prediction). This threshold perfectly separates reliable from unreliable diagnostic regimes across all 16 real LLM–dataset combinations evaluated.

We recommend the following practical protocol: (i) estimate per-subgroup confusion matrices from expert labels; (ii) compute $\max_s \text{score}(s)$; (iii) if > 1, the diagnostic reliably identifies which subgroups are harmed by global correction—apply EC-HTE; (iv) if ≤ 1, the diagnostic is unreliable and practitioners should either default to EC-HTE as a conservative choice or evaluate correction strategies using held-out expert labels.

Y Plug-in Standard Error Validation

A natural concern is whether the plug-in standard error adequately captures estimation uncertainty, particularly the additional variance introduced by confusion matrix estimation. We verify this for the difference-in-means (DIM) estima-

tor via Monte Carlo simulation (K=2, $n_{\text{expert}} \in \{50, 100, 250, 500\}$, 50 replications per budget level). At each budget, we compute the plug-in 95% confidence interval and record empirical coverage. Across all expert budgets, the plug-in SE yields coverage ≥ 0.96 (0.970, 0.960, 0.970, 0.970 for $n_{\text{expert}} = 50, 100, 250, 500$), confirming that the analytic variance estimate is well-calibrated for the DIM estimator. This self-calibration arises because the same noise in $\hat{M}^{-1}$ that inflates point-estimate variance also inflates the plug-in SE, producing conservative intervals that maintain nominal coverage.

CausalForestDML. Under K=2 extreme heterogeneity (π=[0.05, 0.25], n_{expert}=250, $n_{\text{estimators}}$=200, 20 MC runs), EC-HTE plug-in SE achieves 93.8% aggregate coverage—close to the 94–96% oracle coverage (Table 23). The high-error subgroup (s_1, π=0.25) shows slightly lower plug-in coverage (92.5%) than the low-error subgroup (s_0: 95.0%), reflecting the larger M^{-1} amplification of estimation uncertainty in subgroups with higher misclassification rates. By contrast, the DIM plug-in SE achieves ≥95% coverage in the same DGP, confirming that the analytic confusion matrix propagation formula is adequate for simpler estimators.

Full bootstrap SE. A *full* bootstrap—resampling all N observations and re-estimating both the confusion matrix and CausalForest per bootstrap replicate (B=100, $n_{\text{estimators}}$=200, 20 MC runs)—confirms that plug-in SE is already near-nominal under standard hyperparameters. Under K=2 extreme heterogeneity (π=[0.05, 0.25], n_{expert}=250), plug-in SE achieves 93.8% aggregate coverage; the full bootstrap improves this to exactly 95.0% (+1.2pp). Table 22 reports per-subgroup results.

Table 22: CausalForest coverage: plug-in SE vs. full bootstrap SE (B=100, $n_{\text{estimators}}$=200, 20 MC runs). Bootstrap SE is ∼1.15× plug-in.

Subgroup	Plugin Cov.	Boot. Cov.	Oracle Cov.	Plugin SE	Boot. SE
s_0 (π=0.05)	0.950	0.925	0.925	0.059	0.063
s_1 (π=0.25)	0.925	**0.975**	0.875	0.087	0.103
Aggregate	0.938	**0.950**	0.900	0.073	0.083

The high-error subgroup (s_1) benefits most: coverage rises from 92.5% to 97.5% (+5.0pp), because the bootstrap captures the additional variance from M^{-1} amplification. Oracle coverage (90.0%) is consistent with the 94% reported in the oracle decomposition (Table 23), with the difference attributable to MC sampling noise at 20 runs. The computational cost is substantial ($B\times$ CF fits per MC run), but the approach is straightforward when exact nominal coverage is required.

Finite-sample remainder. By first-order matrix perturbation, $\hat{C}_s^{-1} = C_s^{-1} - C_s^{-1} E_s C_s^{-1} + O(\|E_s\|^2)$, where $E_s = \hat{C}_s - C_s$ satisfies $\|E_s\| = O_p(n_{\text{expert}}^{-1/2})$. The remainder in the corrected CATE is therefore $R_n = O_p(n_{\text{expert}}^{-1/2} \cdot \|C_s^{-1}\|^2) = O_p(n_{\text{expert}}^{-1/2} \cdot \kappa(C_s))$, since $\|C_s^{-1}\| = O(\kappa(C_s))$ for stochastic matrices. This motivates the $\kappa > 100$ singularity guard (Section 4).

Y.1 CausalForest Oracle Coverage Decomposition

To isolate the source of CausalForest undercoverage, we compare six estimation strategies that progressively remove sources of uncertainty: (1) Oracle (true Z, no correction), (2) EC-HTE with true C_s, (3) EC-HTE with estimated $\hat{C}_s$ (HB), (4) Global with true $\bar{C}$, (5) Global with estimated $\hat{\bar{C}}$, and (6) Naive. Table 23 reports coverage, |bias|, and RMSE across three DGP configurations (100 MC runs each).

Table 23: CausalForest oracle coverage decomposition. Coverage averaged over subgroups and z-levels.

DGP	Method	Cov. (%)	|Bias|	RMSE
K=4, extreme n=500	Oracle (true Z)	94.5	0.006	0.078
	EC-HTE true C_s	92.5	0.008	0.114
	EC-HTE est. $\hat{C}_s$	93.1	0.015	0.121
	Global true $\bar{C}$	91.3	0.037	0.110
	Global est. $\hat{\bar{C}}$	91.1	0.038	0.111
	Naive	88.3	0.045	0.095
K=2, extreme n=250	Oracle (true Z)	94.0	0.002	0.056
	EC-HTE true C_s	94.0	0.003	0.073
	EC-HTE est. $\hat{C}_s$	92.5	0.004	0.079
	Global true $\bar{C}$	88.5	0.045	0.083
	Global est. $\hat{\bar{C}}$	88.3	0.045	0.085
	Naive	82.8	0.046	0.077
K=2, MF2 DGP γ=(0.10, 0.40)	Oracle (true Z)	96.0	0.077	0.091
	EC-HTE true C_s	91.3	0.103	0.112
	EC-HTE est. $\hat{C}_s$	92.3	0.100	0.111
	Global true $\bar{C}$	95.0	0.107	0.114
	Global est. $\hat{\bar{C}}$	94.3	0.105	0.114
	Naive	77.3	0.120	0.125

Three findings emerge. First, oracle coverage is 94–96% across all DGPs, confirming that CausalForest plug-in SE is well-calibrated when labels are correct. Second, EC-HTE with true C_s

achieves 91–94% coverage—a 0–5pp gap from oracle (configuration-dependent), reflecting M^{-1} SE propagation asymmetry across DGPs rather than confusion matrix estimation error. Third, the gap between true and estimated C_s is $\leq$1.5pp (K=2) and $<$1pp (K=4), confirming that CM estimation uncertainty contributes negligibly at $n_{\text{expert}} \geq 250$.

Z Robustness to Assumption Violations

Z.1 Assumption (A5): Subgroup-Specific Treatment Effects

Corollary 5 (Subgroup-Specific Treatment Effect Heterogeneity)**.** *When $\Delta\tau$ varies across subgroups ($\Delta\tau_s = \tau_s(1) - \tau_s(0)$), the bias of global correction in subgroup s at level z=0 is $+\delta_s \cdot \Delta\tau_s/(1-2\bar{\pi})$, and at level z=1 is $-\delta_s \cdot \Delta\tau_s/(1-2\bar{\pi})$.*

The proof follows directly from Theorem 1 with τ_s replacing τ.

Our main analysis assumes that subgroup misclassification rates δ_s and subgroup CATE magnitudes $\Delta\tau_s$ are independent (Assumption A5). In practice, both may depend on text features that define subgroups, introducing correlation. We test robustness under two configurations with the DIM estimator (K=2, N=5,000, n_{expert}=250, 50 MC replications, $\pi = [0.05, 0.25]$): *aligned* (positive δ_s–$\Delta\tau_s$ correlation: high-error S1 has the larger treatment effect, $\Delta\tau = [0.30, 0.70]$) and *adversarial* (negative correlation: high-error S1 has the smaller effect, $\Delta\tau = [0.70, 0.30]$).

Table 24: Coverage and |bias| under A5 violation (K=2, n_{expert}=250, 50 MC). Target coverage: 0.95.

Config	Method	S0 cov	S1 cov	S0 |bias|	S1 |bias|
Aligned	Oracle	0.94	0.98	0.003	0.003
	Naive	0.93	0.25	0.022	0.174
	Global	0.93	0.82	0.036	0.094
	EC-HTE (EB)	0.95	0.95	0.005	0.017
Adversarial	Oracle	0.94	0.98	0.003	0.003
	Naive	0.90	0.78	0.042	0.074
	Global	0.74	0.93	0.098	0.040
	EC-HTE (EB)	0.94	0.96	0.002	0.007

In the aligned configuration, the $\delta_s \times \Delta\tau_s$ product is maximized in the high-error subgroup, causing naive S1 coverage to collapse to 0.25. Global correction improves S1 but cannot fully correct without subgroup-specific error rates. In the adversarial configuration, global correction overcorrects the low-error subgroup (S0 coverage 0.74). EC-HTE (EB) maintains ≥ 0.94 coverage across both subgroups in both configurations, with bias comparable to the oracle. Results at n_{expert}=500 are consistent (all EC-HTE coverages $\in [0.93, 0.99]$).

Z.2 Assumption (A1): Sensitivity to Outcome-Dependent Misclassification

We assess EC-HTE's robustness when misclassification depends on potential outcomes, violating (A1). We modify the synthetic DGP (K=2, extreme heterogeneity, N=5,000, n_{expert}=250) so that misclassification probability becomes outcome-dependent: $P(\hat{Z} \neq Z \mid Z, S, Y) = \text{clamp}(\pi_s + \gamma \cdot (Y - \mu_Y)/\sigma_Y,\ 0,\ 1)$, where $\gamma \in \{0, 0.05, 0.10, 0.15, 0.20\}$ controls the strength of violation. Table 25 reports results averaged over 100 MC runs.

Table 25: Sensitivity to Assumption (A1) violation. Coverage and |bias| under outcome-dependent misclassification ($\gamma > 0$).

	Coverage (%)			|Bias|		
γ	EC-HTE	Global	Naive	EC-HTE	Global	Naive
0.00	95.5	92.7	87.7	0.060	0.065	0.061
0.05	95.5	93.5	83.8	0.059	0.064	0.064
0.10	95.3	93.8	83.5	0.063	0.064	0.067
0.15	94.8	94.3	79.2	0.065	0.064	0.070
0.20	93.5	94.0	79.5	0.068	0.063	0.073

EC-HTE maintains near-nominal coverage ($\geq$93.5%) across the full γ range, with |bias| increasing only 13.7% from γ=0 to γ=0.2. All correction methods (Global, Stratified MLE, HB) maintain $\geq$90% coverage, indicating that moderate (A1) violations do not catastrophically degrade confusion-matrix-based correction. The naive estimator degrades most rapidly (coverage $-$8.2pp), consistent with outcome-dependent misclassification compounding the uncorrected label noise.

Z.3 Extension to Confounded Treatment Assignment

EC-HTE's confusion matrix correction operates on estimated CATEs regardless of how those estimates are produced. To verify compatibility with observational (non-RCT) designs, we pair EC-HTE with an inverse probability weighted difference-in-means (IPW-DIM) estimator under confounded treatment assignment.

DGP. N=5,000, K=2, $\pi = [0.05, 0.25]$, n_{expert}=250, 100 MC replications. A covariate $X \sim \mathcal{N}(0, 1)$ confounds both treatment and outcome: $T \sim \text{Bernoulli}(\sigma(\beta X))$ with β=0.5 (mod-

erate confounding; propensity range $[0.38, 0.62]$), $Y = 1 + 0.5Z + 0.3X + \tau(Z) \cdot T + \varepsilon, \varepsilon \sim \mathcal{N}(0, 1)$. Treatment effects are $\tau(Z{=}0){=}0$, $\tau(Z{=}1){=}0.3$. The IPW-DIM estimator uses *oracle propensity* $e(X) = \sigma(\beta X)$, with propensity scores clipped to $[0.05, 0.95]$.

Table 26: EC-HTE under confounded treatment assignment (IPW-DIM with oracle propensity, $\beta{=}0.5$, 100 MC).

Method	Coverage	RMSE	|Bias|
Oracle (true Z + IPW)	0.943	0.061	0.048
HB EC-HTE	0.950	0.086	0.068
Global	0.930	0.087	0.071
Strat MLE	0.950	0.087	0.069
Naive	0.863	0.083	0.066

EC-HTE achieves 95.0% coverage, within 0.7pp of the oracle (94.3%), confirming that confusion matrix correction and propensity weighting compose without interference. The improvement is most pronounced in the high-error subgroup ($\pi{=}0.25$): EC-HTE coverage is 96.0% vs. naive 78.5% ($+17.5$pp). Global coverage (93.0%) falls below the correction methods (95.0%), consistent with the cost of ignoring subgroup-level error heterogeneity even in the confounded setting.

Sensitivity to confounding strength and propensity estimation. We sweep $\beta \in \{0.25, 0.5, 1.0, 2.0\}$ and compare oracle propensity ($e(X) = \sigma(\beta X)$) with estimated propensity (logistic regression fit on (X, T)), yielding 8 configurations (100 MC each). Table 27 reports HB EC-HTE and naive coverage.

Table 27: EC-HTE coverage across confounding strengths and propensity sources. The logistic regression propensity model is correctly specified; propensity misspecification is an orthogonal concern addressed by the IPW literature.

β	Propensity	HB Coverage	Naive Coverage
0.25	Oracle	0.963	0.848
0.25	Estimated	0.973	0.845
0.50	Oracle	0.950	0.863
0.50	Estimated	0.958	0.858
1.00	Oracle	0.958	0.898
1.00	Estimated	0.960	0.883
2.00	Oracle	0.945	0.858
2.00	Estimated	0.948	0.858

EC-HTE maintains coverage $\geq$94.5% across all 8 configurations, including strong confounding ($\beta{=}2.0$, propensity range $[0.12, 0.88]$). The oracle–estimated gap is $\leq$1pp, confirming that propensity estimation error does not meaningfully degrade EC-HTE when the propensity model is correctly specified. This validates that EC-HTE is robust to confounded treatment assignment when the propensity model is correctly specified.

Z.4 Empirical Comparison with Frazis–Loewenstein Correction

We empirically compare EC-HTE's confusion matrix inversion against the scalar attenuation correction of Frazis and Loewenstein (2003) (F-L). The F-L correction rescales the observed estimate as $\hat{\tau}_{\text{FL}} = \hat{\tau}_{\text{obs}}/(1 - 2\bar{\pi})$, where $\bar{\pi}$ is the (global or per-subgroup) misclassification rate. We test two variants: F-L *scalar* (single global $\bar{\pi}$) and F-L *per-subgroup* (π_s-specific rescaling). Setup: $K{=}2$, $N{=}5{,}000$, $n_{\text{expert}}{=}250$, $\pi{=}[0.05, 0.25]$ (extreme heterogeneity), $\tau(z{=}0){=}0$, $\tau(z{=}1){=}0.3$, 100 MC runs.

Table 28: F-L correction vs. EC-HTE under heterogeneous misclassification ($\Delta\pi{=}0.20$). Both F-L variants produce *higher* aggregate bias than the uncorrected estimator.

Method	|Bias|	RMSE	Coverage
Oracle	0.045	0.056	0.943
Naive	0.062	0.077	0.858
Global (CM inv)	0.064	0.081	0.938
HB EC-HTE	0.064	0.086	**0.953**
F-L Scalar	0.091	0.116	0.850
F-L Per-Subgroup	0.112	0.158	0.838

Both F-L variants produce *higher* bias and RMSE than the naive estimator (F-L per-subgroup $|\text{bias}|{=}0.112$ vs. naive 0.062), demonstrating that scalar attenuation cannot substitute for confusion matrix inversion under subgroup-heterogeneous misclassification. The per-subgroup bias direction reveals the mechanism: F-L scalar over-corrects the low-error subgroup (s_0: mean bias $= +0.073$ vs. naive $+0.005$, a $14.6\times$ amplification), while F-L per-subgroup dramatically over-corrects the high-error subgroup (s_1: mean bias $=+0.160$). EC-HTE achieves near-zero per-subgroup bias (s_0: $+0.006$, s_1: $+0.001$) with nominal coverage (95.3%). The root cause is structural: F-L's $1/(1{-}2\pi)$ scaling amplifies each z-level estimate independently, whereas CM inversion simultaneously redistributes the cross-contamination between misclassified and correctly classified observations.

Z.5 End-to-End Validation with Asymmetric Confusion Matrices

We validate EC-HTE under strongly asymmetric confusion matrices (FPR/FNR ratio up to 7:1), simulating the error profile of GPT-4.1 on Civil Comments. Setup: K=2, N=5,000, n_{expert}=250; s_0: FPR=0.05, FNR=0.02 (low error); s_1: FPR=0.38, FNR=0.054 (7:1 ratio); 100 MC runs. We compare EC-HTE using the full 2×2 asymmetric CM (Remark 2) against the symmetric assumption (FPR=FNR=π_s).

Table 29: EC-HTE under asymmetric misclassification (FPR/FNR up to 7:1). The asymmetric formula improves coverage by +1.5pp and RMSE by 11.5%.

Method	|Bias|	RMSE	Coverage
Oracle	0.045	0.056	0.943
EC-HTE (asymmetric)	0.057	0.074	**0.945**
Naive	0.059	0.073	0.870
Global (asymmetric)	0.061	0.076	0.915
EC-HTE (symmetric)	0.062	0.083	0.930
Global (symmetric)	0.063	0.078	0.915

EC-HTE with the asymmetric formula achieves 94.5% coverage versus 93.0% under the symmetric assumption (+1.5pp), with RMSE improved from 0.083 to 0.074 (−11.5%). The improvement concentrates in the high-error subgroup (s_1): asymmetric coverage is 93.0% vs. symmetric 91.0%. Practitioners should estimate full 2×2 confusion matrices rather than assuming symmetric misclassification, particularly when FPR and FNR differ substantially.

Z.6 Complementarity with LLM Calibration

Temperature or Platt scaling (Guo et al., 2017) can reduce per-subgroup misclassification rates, but does not eliminate heterogeneity $\Delta\pi$. We simulate calibration by shrinking per-subgroup rates from π=$[0.05, 0.25]$ ($\Delta\pi$=0.20) to π=$[0.08, 0.15]$ ($\Delta\pi$=0.07), testing whether EC-HTE remains necessary post-calibration. Setup: K=2, N=5,000, n_{expert}=250, 100 MC runs.

Calibration alone improves naive coverage from 84.0% to 91.3% but does not reach the nominal 95%. Calibrated + EC-HTE achieves the highest coverage (96.5%), demonstrating that calibration and EC-HTE address orthogonal error sources: calibration reduces per-subgroup π_s while EC-HTE corrects the residual δ_s-driven bias from heterogeneity that persists post-calibration. Propensity estimation robustness under model misspecification is an orthogonal concern addressed by the IPW literature; EC-HTE as a downstream correction layer does not alter IPW behavior.

Table 30: Calibration reduces $\Delta\pi$ but does not eliminate it; EC-HTE corrects the residual bias. Best coverage is achieved by combining both.

Method	|Bias|	RMSE	Coverage
Oracle	0.043	0.053	0.963
Calibrated + EC-HTE	0.055	0.069	**0.965**
Calibrated + Global	0.055	0.068	0.963
Calibrated + Naive	0.051	0.064	0.913
Uncalibrated + EC-HTE	0.064	0.084	0.938
Uncalibrated + Global	0.069	0.085	0.920
Uncalibrated + Naive	0.063	0.079	0.840

Algorithm 1 EC-HTE: Error-Corrected Heterogeneous Treatment Effects

Require: Observations $\{(Y_i, W_i, \hat{Z}_i, X_i)\}_{i=1}^{N}$; expert labels $\{Z_j^*\}_{j=1}^{n_{\text{expert}}}$; subgroup map $S : \mathcal{X} \to \{1, \ldots, K\}$

Ensure: Corrected CATE estimates $\hat{\tau}(z)$ with variance-adjusted CIs

Preconditions: verify $n_{\text{expert}} \times P(Z_{\text{minority}})/K \geq 25$ and $\max_s(\text{FPR}_s, \text{FNR}_s) < 0.5$; if either fails, use global correction

Step 1: Estimate per-subgroup confusion matrices (HB)

for $s = 1, \ldots, K$ **do**

  $C_s[j, \cdot] \sim \text{Dir}(\alpha_0 \cdot \mathbf{e}_j)$, $\alpha_0 \sim \text{Gamma}(3, 1)$

  Gibbs sampler (4 chains, 4000 iter) on $\{(\hat{Z}_j, Z_j^*) : S_j = s\}$

  $\hat{C}_s \leftarrow$ posterior mean; verify $\hat{R} < 1.01$

end for

Step 2: Correct observed outcome moments

for $s = 1, \ldots, K$ **do**

  if $\kappa(\hat{C}_s) > 100$ **then** $\hat{C}_s \leftarrow \bar{C}$ ▷ singularity guard

  end if

  $\tilde{\boldsymbol{\mu}}_s \leftarrow \hat{C}_s^{-1} \hat{\boldsymbol{\mu}}_s$

end for

Step 3: CausalForestDML with variance-adjusted SEs

Train CausalForest (honest splitting) on corrected labels

$\text{se}_{\text{adj},i} \leftarrow \sqrt{\text{diag}(\hat{C}_{S_i}^{-1} \, \text{diag}(\text{se}_i^2) \, \hat{C}_{S_i}^{-\top})}$

return $\hat{\tau}(z)$ with 95% CIs from se_{adj}